

Cellular Stress Responses And Cell Death Decisions Laboratory Protocol

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August 16, 2024

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1 Pipette Calibration Test

1.1 Introduction

Pipettes are crucial instruments in biological laboratories for precise measurement and transfer of liquid volumes. Accurate pipetting is essential for reliable experimental results, and both the pipettes and the users must be thoroughly calibrated and tested. This experiment aims to assess the accuracy and precision of pipetting performed by two individuals, referred to as "t" and "l", using different pipettes and volumes of water.

1.2 Objective

The objective of this calibration test was to evaluate the accuracy and precision of measurements made by two individuals using various pipettes. By analyzing the data from each individual and pipette, we aimed to assess the performance and identify any discrepancies that could impact experimental outcomes.

1.3 Materials and Methods

The experiment was conducted following standard laboratory procedures. Detailed information about the procedure is available in the laboratory script.

1.4 Results

The measurements for individuals "t" and "l" are detailed in the following links: [t-measurement-data](#) and [l measurement-data](#).

For each pipette and volume tested, the average of the three measurements was calculated to determine the mean dispensed volume. The standard deviation was also computed to assess the consistency of the pipette's performance. The mean dispensed volume was then compared to the nominal volume values to evaluate accuracy.

The following steps were undertaken to analyze the calibration data:

Identify Outliers: The Z-score method was employed to detect outliers in the measurements. Data points with Z-scores exceeding 3 were considered outliers and were flagged for further review. However, no outliers were detected in this study, allowing for a straightforward analysis.

Calculate Deviations: The deviation of each pipette's mean measurement from the nominal volume was calculated to assess accuracy. This was done using the formula:

$$\text{Deviation} = \text{Mean Dispersed Volume} - \text{Nominal Volume}$$

Relative Error (%): The relative error was computed to evaluate the precision of the measurements:

$$\text{Relative Error (\%)} = \left(\frac{\text{Deviation}}{\text{Nominal Volume}} \right) \times 100$$

Coefficient of Variation (CV %): The coefficient of variation was calculated to measure the pipetting precision:

$$\text{CV (\%)} = \left(\frac{\text{Standard Deviation}}{\text{Mean Dispensed Volume}} \right) \times 100$$

Expanded Uncertainty: The expanded uncertainty was estimated to account for the confidence interval, with a coverage factor (k) of 2:

$$\text{Expanded Uncertainty} = 2 \times \text{Standard Deviation}$$

The results of these analyses are presented in Figure 1.

	Pipette	Volume	Mean (g)	Standard Deviation (g)	Nominal Volume (μL)	Deviation (g)	Relative Error (%)	CV (%)	Expanded Uncertainty (g)
"t" individual analysis	0	p10	1μ	0.0010	1.0	0.0000	0.0000	10.0000	0.0002
	1	p10	5μ	0.0048	5.0	-0.0002	-4.6667	2.4224	0.0002
	2	p10	10μ	0.0097	10.0	-0.0003	-2.6667	1.1863	0.0002
	3	p200	20μ	0.0194	20.0	-0.0006	-3.1667	0.5962	0.0002
	4	p200	100μ	0.0983	100.0	-0.0017	-1.7333	0.2561	0.0005
	5	p200	200μ	0.1977	200.0	-0.0023	-1.1500	0.0506	0.0002
	6	p1000	100μ	0.0982	100.0	-0.0018	-1.7667	0.5877	0.0012
	7	p1000	500μ	0.4936	500.0	-0.0064	-1.2800	0.2301	0.0023
	8	p1000	1000μ	0.9875	1000.0	-0.0125	-1.2533	0.2149	0.0042
	Pipette	Volume	Mean (g)	Standard Deviation (g)	Nominal Volume (μL)	Deviation (g)	Relative Error (%)	CV (%)	Expanded Uncertainty (g)
"l" individual analysis	0	p20	2μ	0.0058	2.0	0.0038	190.0000	122.4502	0.0142
	1	p20	10μ	0.0094	10.0	-0.0006	-6.3333	5.8800	0.0011
	2	p20	20μ	0.0798	20.0	0.0598	298.8333	131.6232	0.2100
	3	p200	20μ	0.0197	20.0	-0.0003	-1.3333	1.9186	0.0008
	4	p200	100μ	0.0995	100.0	-0.0005	-0.5333	0.5159	0.0010
	5	p200	200μ	0.2021	200.0	0.0021	1.0333	1.2572	0.0051

Figure 1: Tables of analyzed data for individuals "t" (top table) and "l" (bottom table). Each table includes measurements for different pipettes and volumes, showing the mean and standard deviation nominal volumes, deviation, relative errors, coefficient of variation(CV) and expected uncertainty of three measurements.

1.5 Discussion

The calibration test provided important insights into both the performance of the pipettes and the pipetting skills of individuals "t" and "l":

- **Individual t:** Demonstrated a high level of accuracy and precision across all volumes and pipettes. For example, the pipette *p10* at 1 μL had a deviation of 0.0000 g from the nominal value, and a very low coefficient of variation (CV) of 10.00%. Similar patterns were observed for larger volumes and other pipettes, with relative errors generally below 5%. The low standard deviations and tight error margins indicate that both the pipette performance and individual t's technique are highly reliable, suggesting proficiency in handling and consistent pipette calibration.

- **Individual l:** Exhibited more variability in measurements, particularly at lower volumes. The pipette *p20* at 2 μL showed a substantial deviation of 0.0038 g, with a high relative error of 190.00% and an expanded uncertainty of 0.0142 g. This significant deviation suggests possible issues with pipette handling or calibration. Additionally, the *p20* at 20 μL showed a

deviation of 0.0598 g and a CV of 131.62%, indicating a lack of precision and consistency in measurements. For the *p200* pipette, the errors were somewhat reduced but still noticeable, particularly at the higher volumes, where more consistent errors suggest either technique issues or the need for recalibration.

The test also demonstrated that the pipettes themselves are generally reliable, as most observed variations in measurements were likely due to individual performance rather than equipment malfunction. Regular calibration checks combined with targeted training for individuals exhibiting variable performance will help enhance overall accuracy and reliability in future experiments.

1.6 Conclusion

The pipette calibration test underscores the critical need for both accurate pipette performance and proficient pipetting technique.

- **Individual t** showed exemplary performance, with measurements closely aligning with the nominal volumes and minimal deviations. This indicates that both the pipette and the pipetting technique are well-calibrated and reliable.

- **Individual l's** results highlighted a need for improvement, particularly in achieving accuracy with lower volumes. The substantial deviations observed suggest that individual l would benefit from additional training and possibly further calibration of the pipettes.

Ensuring precise pipetting through regular calibration and refining technique is essential for obtaining reliable and reproducible experimental results. For a detailed breakdown of calculations and additional data, please refer to the comprehensive analysis available at Pipette Calibration Test Computations.

1.7 References

J. Blues, D. Bayliss, M. Buckley. "The calibration and use of piston pipettes".
Scioninstruments.com. The importance of laboratory equipment calibration.
Integra-biosciences.com. Calibration check

2 Continuous Cell Culture

2.1 Introduction

Cell culture is a fundamental method in biological research that enables the in vitro study of cell behavior under controlled conditions. In continuous cell culture, cell lines, such as immortalized HeLa cells, which can be passed on indefinitely, are maintained and propagated over a long period of time. However, as the number of passages increases, the likelihood of genetic drift and contamination may increase, potentially impacting experimental results.

2.2 Objective

The objective of this experiment is to maintain and monitor continuous HeLa cell cultures over a period of 12 days. This includes regular cell counting, freezing and thawing cells, and testing for contamination. We aim to ensure healthy cell growth and assess our techniques for maintaining cell cultures without antibiotics.

2.3 Materials and Methods

Materials:

- HeLa cells
- Cell culture flasks (T25, T75)
- RPMI 1640 medium with 10% Fetal Bovine Serum (FBS) and 2 mM Glutamine
- Phosphate-Buffered Saline (PBS)
- Trypsin/EDTA solution
- Trypan Blue solution
- 15 ml blue cap tubes
- 1 ml plastic pipettes and pipette tips
- Neubauer hemocytometer
- Cryo vials
- Mr. Frosty box
- Freezing medium (80% FBS + 20% DMSO)
- Humidified incubator with 5% CO_2 at 37°C
- Microscope
- 70% Ethanol

Methods

The experiment was performed according to the laboratory script. Detailed information on the procedure can be found in the laboratory script for this experiment.

2.4 Results

2.4.1 Counting cells

During the experiment, we occasionally counted cells (data is available here) for various reasons:

Ensuring optimal cell density: Accurate cell counting helps maintain optimal cell density, which is critical for cell health, proliferation, and experimental consistency. Too high a cell density can lead to nutrient depletion, waste accumulation, and altered cell behavior, while too low a cell density can result in poor cell growth and inconsistent results.

Assessment of cell viability: Using a viability dye such as trypan blue, cell counting allows us to assess the ratio of live to dead cells, providing important information about the condition of the cell culture.

Adjusting cell seeding density: Knowing the cell concentration allows appropriate adjustment of seeding density for new experiments and ensures that cells are plated at an optimal density for their growth and experimental requirements.

Standardization of cell numbers for experiments: In some experiments, we counted the cells to determine the required volume of cell suspension. The following formula is used for this purpose:

$$\text{Counted cells (average of four squares)} \times 2 \times 10^4$$

We calculated the cell concentration (cells/ml) using this formula. This formula takes into account the dilution factor, 2, and the volume factor, 10^4 , to convert the average number of cells counted into the number of cells per milliliter.

2.4.2 Freezing and Thawing Cells

To ensure the viability of the HeLa cells for future experiments, we cryopreserved surplus cells (day 5) using a controlled freezing protocol (detailed steps can be found in the lab protocol). Below you will find the results and the exact steps we followed:

Based on the cell count result of 26.5 cells per $10\mu l$, we calculated the cell concentration as follows:

$$\text{Counted cells} = 26.5 \times 2 \times 10^4 = 530000 \text{ cells/ml}$$

To determine the volume of 530,000 *cells/ml* suspension required to obtain 3 million cells in 500 μl , we performed the following calculation:

$$\text{Volume needed} = \frac{3 \times 10^6 \text{ cells}}{530,000 \text{ cells/ml}} \approx 5.66 \text{ ml}$$

We then took 5.66ml of the original cell suspension and centrifuged this volume to concentrate the cells. After centrifugation, the cell pellet was resuspended in 500 μl of medium to achieve a final cell concentration of approximately 3 million cells in this volume.

We then added 500 μl of the freezing medium to the cryovial. The vial was then carefully inverted 2 – 3 times to mix the contents thoroughly. Finally, the cryovial was placed in a Mr. Frosty box to ensure a controlled freezing rate. This procedure allowed us to effectively cryopreserve the HeLa cells and maintain their viability for future experiments.

The cryopreserved cells were thawed on Day 11 and subsequently cultured (detailed steps can be found in the lab protocol). The culture was checked the following day and no unusual observations were made. The cells remained healthy, exhibited good and even distribution, and showed no signs of contamination.

2.4.3 Contaminations during Cell Culture

During the course, I regularly checked a 25 cm^2 flask used for cell culturing every two days and found no signs of contamination. The cells generally grew well, though sometimes they were unevenly distributed due to incomplete mixing during the transfer process.

To test for contamination, I deliberately exposed the actively growing cell culture (I did not contaminate the cell culture from the thawed cells!) to lab dust, which may contain bacteria (see figure 2).

The next day (day 13), I noticed minimal bacterial activity, mainly bacilli-shaped bacteria. Possible reasons for the low bacterial presence include:

Stable Environment: The controlled conditions of the 37 $^{\circ}\text{C}$, 5% CO_2 incubator might have inhibited bacterial growth.

Natural Factors: Components in the cell culture medium could have limited bacterial growth.

Laboratory Tidiness: The overall cleanliness and strict sterile techniques practiced in the laboratory may have contributed to the low levels of contamination.

Contamination Method: The way I contaminated the sample might not have been the most effective. I simply threw some dust into the sample. A more effective method could have been to dissolve the dust in a liquid, such as water, and then use that solution to contaminate the sample.

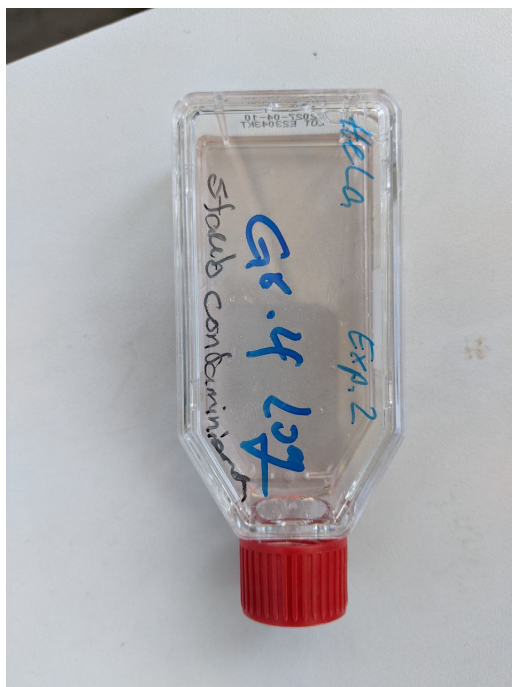


Figure 2: HeLa cell culture contaminated with laboratory dust

2.5 Discussion

The results of the continuous cell culture experiment gave us valuable insights. We were able to maintain HeLa cells for 12 days without antibiotics, showing that our sterile techniques were effective. Regular cell counting helped us keep the cell density at optimal levels, which is important for healthy cell growth.

The freezing and thawing of cells worked well. The cryopreserved cells stayed viable and grew healthily after thawing. This shows that our methods for freezing and storing cells are reliable.

Testing for contamination was also informative. Deliberately contaminating the culture with dust resulted in minimal bacterial growth. This suggests that the controlled environment of the incubator and the components in the growth medium helped prevent bacterial contamination. Our lab's cleanliness and good sterile practices likely contributed to this outcome.

Overall, the experiment highlights the importance of maintaining sterile conditions and regularly monitoring cell cultures. These practices help ensure reliable results in biological research.

2.6 Conclusion

Throughout the experiment, we successfully maintained HeLa cell cultures for 12 days without antibiotics, focusing on sterile techniques to avoid contamination. Regular monitoring

of cell density and morphology ensured healthy cell growth and constant experimental conditions.

Cell counting served as a guide for our procedures and helped us to effectively control cell seeding and freezing. Cryopreservation on Day 5 and subsequent thawing on Day 11 showed good cell viability and growth, demonstrating the robustness of the HeLa cell line and their ability to be frozen and preserved for a period of time.

In the contamination controls on day 13, intentional exposure to laboratory dust showed minimal bacterial activity, likely due to the controlled environment of our incubator and the components in our growth medium that inhibit bacterial growth.

This experiment underscores the importance of careful cell culture practices and proactive contamination control in maintaining reliable research results in biological studies.

2.7 References

Science Direct. Continuous cell culture

Thermofisher.com. Introduction to Cell Culture

3 Gene Function Analysis (CRISPR/Cas9)

3.1 Introduction

Analyzing gene function is an essential part of understanding how specific proteins contribute to cellular processes. In this experiment, we focus on the HeLa cell line, a human cervical adenocarcinoma cell line known for its rapid doubling time and genetic stability. By using techniques to delete or knock out specific proteins, we can observe changes in cell behavior and signaling pathways. An effective method for producing knockout cell lines is the CRISPR/Cas9 technique, which allows precise editing of the genome. The aim of this study is to understand the role of TRAIL-R1 (DR4) and TRAIL-R2 (DR5) receptors in TRAIL-induced apoptosis signaling by generating HeLa cell lines lacking these receptors.

3.2 Objective

The aim of this experiment was to investigate the contribution of TRAIL-R1 (DR4, TNFRSF10a) and TRAIL-R2 (DR5, TNFRSF10b) to TRAIL-induced apoptosis in HeLa cells. We attempted to disrupt the TNFRSF10a or TNFRSF10b genes using the CRISPR/Cas9 method. After HeLa cells were transfected with specific vector plasmids, the knockout was not verified on these cells directly due to time constraints. Instead, pre-established knockout cell lines, provided by the laboratory, were used for Western blot analysis to confirm the disruption of TRAIL-R1 and TRAIL-R2. Subsequently, the sensitivity of these knockout cells to TRAIL-induced apoptosis was compared to that of the parental HeLa cells by analyzing the cleavage of the caspase substrate PARP.

3.3 Materials and Methods

The experiment was performed according to the laboratory script. Detailed information on the procedure can be found in the laboratory script for this experiment.

3.4 Results

3.4.1 Seed Cells

Based on the cell count result of ≈ 41 cells per $10 \mu l$, we calculated the cell concentration as follows:

$$\text{Counted cells} = 41 \times 2 \times 10^4 = 820000 \text{ cells/ml}$$

We prepared a suspension for three wells (one well extra!), which means a total of 9×10^5 cells in 7.5 ml suspension. To calculate the volume of cell culture needed, we performed the following calculation:

$$\text{Cell Culture volume needed} = \frac{9 \times 10^5 \text{ cells}}{820000 \text{ cells/ml}} \approx 1.1 \text{ ml}$$

$$\text{Medium volume needed} = 7.5 \text{ ml}(\text{Suspension needed}) - 1.1 \text{ ml} = 6.4 \text{ ml}$$

Then, the plates were placed in an incubator ($CO^2 = 5\%$, $t = 37^\circ C$) to cultivate the cells for the remainder of the experiment.

3.4.2 Cell Transfection

The following day, we transfected the seeded cells from one of the wells (the other well served as a non-transfected, non-fluorescent control) with the CRISPR/Cas9 construct: CRISPR/-Cas9 construct containing gRNA 8, Cas9 enzyme, and DR receptor 2, DR5/3-GFP-PX458 plasmid.

3.4.3 Fluorescence-assisted cell sorting (FACS)

After transfection with the CRISPR/Cas9 construct, GFP-positive HeLa cells were identified using an EVOS epi-fluorescence microscope (figure 3). These cells were then harvested alongside non-transfected control cells for FACS calibration. The procedure involved washing cells with PBS, trypsinizing them, and preparing a cell suspension in FACS-PBS. Following centrifugation and filtration to remove clusters, the cells were sorted using FACS based on GFP fluorescence intensity.

Successfully transfected cells were seeded individually into two 96-well plates, ensuring optimal growth conditions in FACS-medium supplemented with conditioned medium. The plates were incubated overnight to allow cell attachment. On the following day, wells with viable, dividing cells were identified and marked. These successful clones were monitored and maintained throughout the course, with medium changes as needed to support colony growth.

Each single cell has the potential to proliferate into a colony of over 16,000 cells within 15 days. Although the colonies were monitored, the transfer of colonies to larger culture vessels was carried out by the laboratory personnel. Ultimately, the aim is to expand successful clones for further experiments, including verification of gene disruption and potential cryopreservation for future use.

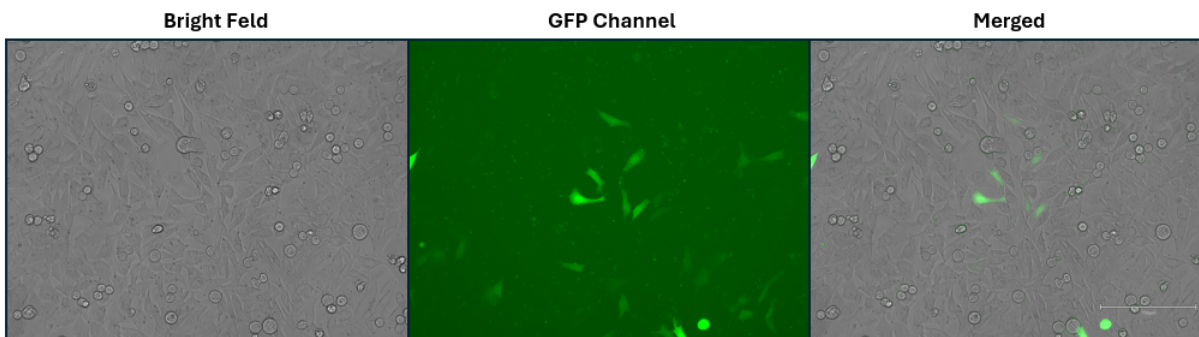


Figure 3: Detection of transfected HeLa cells using the EVOS epi-fluorescence microscope. The left image shows the cell culture in the well under bright field illumination. The middle image displays the same well under fluorescence illumination, highlighting GFP-positive transfected cells in bright green. The right image is a merged view of the bright field and fluorescence images, making the transfected cells more visible. The scale bar corresponds to 150 μ m.

3.4.4 Seeding of a (previously generated) stable knockout cell line

In this part of the experiment, we used a previously generated stable knockout cell line, HeLa DR5 KO (that were given to us), instead of our own samples. The reason/s behind this: because generating a stable knockout cell line involves multiple steps that are time-consuming and have a high failure rate. These steps include designing and purifying a suitable CRISPR/Cas9 plasmid, successful transfection, isolating a single transfected cell, and growing a viable cell population. The process can take weeks, and the stress of transfection and sorting can lead to cell death or senescence, preventing us from completing the experiment within the available timeframe.

To seed the cells:

We prepared 14ml of cell suspension for the *DR5* knockout cells:

We counted an average of 25 cells per 10 μ l, giving a concentration of $25 \times 2 \times 10^4 = 500,000$ cells/ml.

We needed 12 ml for the 6 wells and 2 ml extra for adjustments. We used 5.6 ml of cell culture (500,000 cells/ml) and added 8.4 ml of medium to make up the 14 ml.

We also prepared 14 ml of cell suspension for the wild-type cells:

We counted an average of 27 cells per 10 μ l, giving a concentration of $27 \times 2 \times 10^4 = 540,000$ cells/ml.

We needed 12 ml for the 6 wells and 2 ml extra for adjustments. We used approximately 5.2 ml of cell culture (540,000 cells/ml) and added approximately 8.8 ml of medium to make up the 14 ml.

For both cell lines, we seeded 400,000 cells in 2 ml *RPMI*/10% FBS/PenStrep per well in all 6 wells of the 6-well plates.

We labeled the plates with the cell line names to avoid mix-ups and incubated the cells overnight to allow them to attach.

3.4.5 Treatment

In this part of the experiment, we evaluated the resistance of prepared cell lines (HeLa-wt and HeLa DR5-KO) to TRAIL-induced apoptosis. The disruption of the TNFRSF10b gene depletes cells of the TRAIL-R2 receptor, increasing resistance to TRAIL-induced apoptosis.

Two 6-well plates were used for the experiment (prepared on Day 6):

Plate 1: HeLa TRAIL-R2 knockout cells (KO).

Plate 2: Wild-type HeLa cells (wt), serving as the control.

Four wells in each plate were chosen for the experiment: three of them were treated with different concentrations of TRAIL, and one of them was untreated as a control. The treatment scheme for the wells was as follows:

Plate 1 (HeLa TRAIL-R2 KO cells):

1) Untreated control 2) 10 ng/ml TRAIL 3) 50 ng/ml TRAIL 4) 100 ng/ml TRAIL

Plate 2 (Wild-type HeLa cells):

The same treatment concentrations as Plate 1 to serve as the control for comparison.

To prepare the TRAIL dilutions, the following steps were followed from the stock concentration of 100 µg/ml:

Prepare 1 µg/ml TRAIL by mixing 10 µl of TRAIL stock with 990 µl of medium. Dilute further to achieve the desired concentrations:

For 10 ng/ml TRAIL, mix 45 µl of the 1 µg/ml TRAIL with 4,455 µl of medium.

For 50 ng/ml TRAIL, mix 225 µl of the 1 µg/ml TRAIL with 4,275 µl of medium.

For 100 ng/ml TRAIL, mix 450 µl of the 1 µg/ml TRAIL with 3,550 µl of medium.

The plates were then incubated for 24 hours.

The next day, we observed that HeLa TRAIL-R2 knockout (KO) cells and wild-type (wt) HeLa cells responded differently to TRAIL treatment. The KO cells were more resistant to cell death compared to the wt cells at all TRAIL concentrations. While the wt cells showed a clear decrease in viability with higher TRAIL doses, the KO cells remained more viable even at higher TRAIL concentrations. This shows that disrupting the TNFRSF10b gene in KO cells makes them more resistant to TRAIL-induced apoptosis.

3.4.6 Cell Harvesting and Protein Quantification

On Day 8, we harvested the cells to prepare whole-cell lysates (WCL) and determined the protein concentration using the Bradford method. The cells were collected from the wells, including both attached and floating cells, and transferred into labeled tubes. After washing with PBS and trypsinizing, the cells were centrifuged and lysed to obtain the WCL. To measure the protein concentration, we used the Bradford method with BSA standards to create a reference curve. The absorbance at 595 nm was measured, and the protein concentration of each lysate was calculated based on this reference.

3.4.7 Western blotting

To analyze more the protein expression levels in whole cell lysates of HeLa cells, both wild-type and TRAIL-R2 knockout (KO) cells, after treatment with varying concentrations of TRAIL, western blot technique is used, we aim to determine the impact of the TNFRSF10B gene knockout on the resistance of HeLa cells to TRAIL-induced apoptosis. This will be achieved by comparing the protein levels of the target proteins, DR5, PARP, and GAPDH, between the treated and untreated samples.(figure 4)

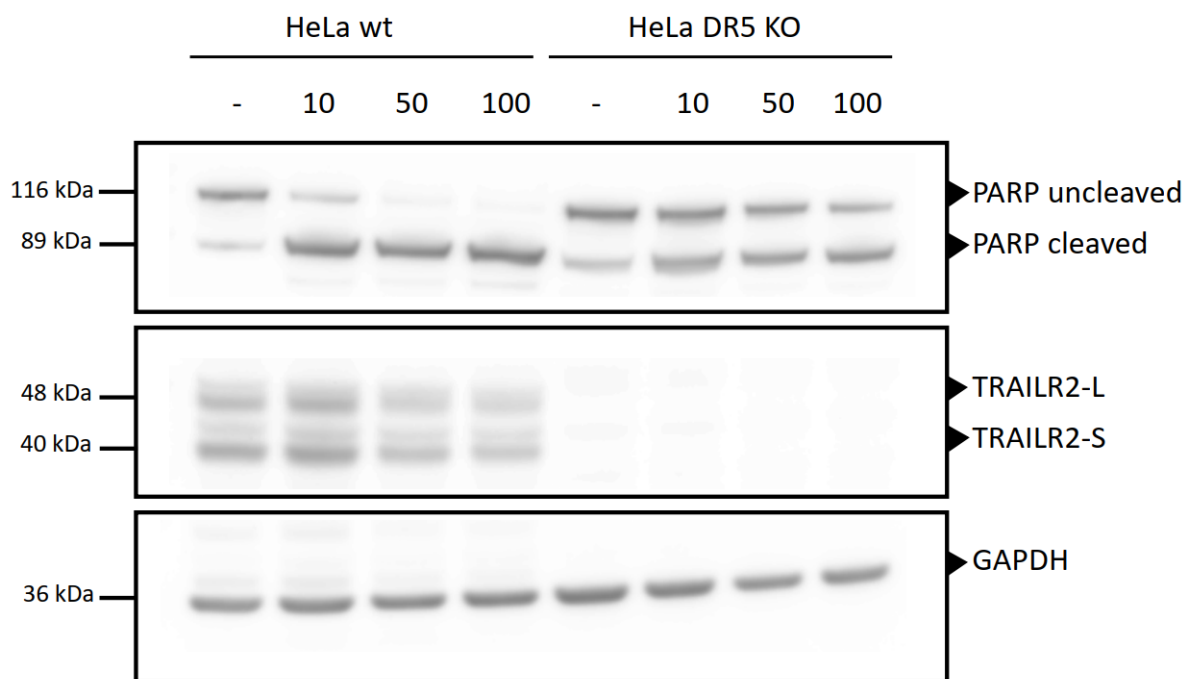


Figure 4: Western blot analysis comparing PARP cleavage, TRAIL-R2 expression, and GAPDH as a loading control in wild-type (HeLa wt) and TRAIL-R2 knockout (HeLa DR5 KO) HeLa cells. Cells were treated with increasing concentrations of TRAIL (0, 10, 50, 100 ng/ml). The blot shows that PARP cleavage is significantly higher in wild-type cells, indicating increased apoptosis, while TRAIL-R2 knockout cells exhibit reduced PARP cleavage and higher levels of uncleaved PARP, highlighting their increased resistance to TRAIL-induced apoptosis.

Increased Resistance in KO Cells:

The KO cells show higher viability at all TRAIL concentrations compared to WT cells. This suggests that knocking out the TNFRSF10b gene (encoding TRAIL-R2) increases resistance to TRAIL-induced apoptosis.

Cleaved PARP as an Apoptotic Marker:

The percentage of cleaved PARP increases with higher TRAIL concentrations in both WT and KO cells, figure 5, indicating that TRAIL induces apoptosis in a dose-dependent

manner. However, the proportion of cleaved PARP is significantly higher in WT cells at higher TRAIL concentrations (50 and 100 ng/ml), suggesting that WT cells undergo more apoptosis compared to KO cells.

Uncleaved PARP:

Correspondingly, the percentage of uncleaved PARP decreases with increasing TRAIL concentrations in both cell types. WT cells show a dramatic decrease in uncleaved PARP with increasing TRAIL, whereas KO cells retain higher levels of uncleaved PARP, indicating reduced apoptosis.

Cell Viability:

The viability data show that KO cells have higher resistance to TRAIL-induced cell death, maintaining higher viability even at the highest TRAIL concentrations (50 and 100 ng/ml).

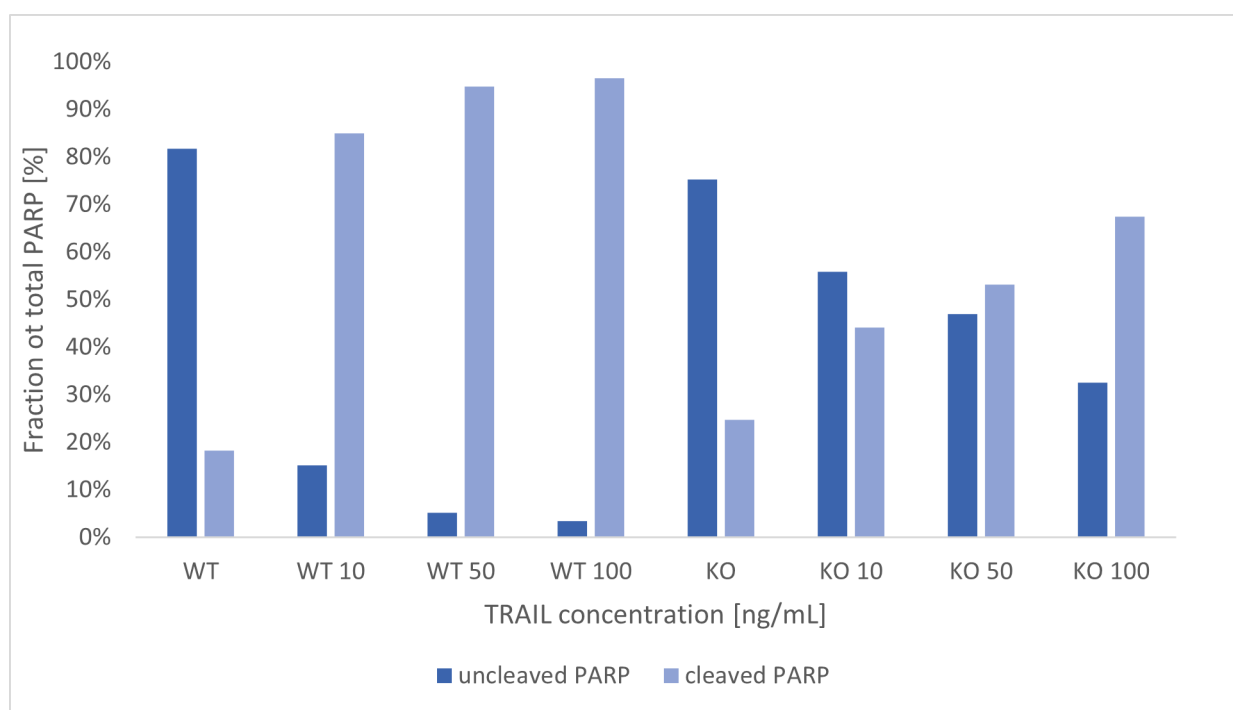


Figure 5: The bar diagrams illustrate the protein expression levels of PARP in whole cell lysates of HeLa cells, comparing wildtype (WT) and TRAIL-R2 knockout (KO) cells after a 24-hour treatment with varying concentrations of TRAIL. The X-axis represents different treatment conditions, including untreated control samples (WT and KO), and samples treated with 10 ng/ml, 50 ng/ml, and 100 ng/ml TRAIL (labeled as WT10, WT50, WT100 for wildtype, and KO10, KO50, KO100 for knockout cells). The Y-axis indicates the fraction of total PARP protein expression. This comparison allows for an assessment of the impact of TRAIL and the absence of TRAIL-R2 on PARP levels, with the WT and KO bars revealing differences in protein expression between treated and untreated cells across the range of TRAIL concentrations.

3.5 Discussion

In this experiment, the role of TRAIL-R1 (DR4) and TRAIL-R2 (DR5) receptors in TRAIL-induced apoptosis in HeLa cells was investigated using CRISPR/Cas9 technology to disrupt the TNFRSF10a and TNFRSF10b genes. The findings revealed that knocking out the TNFRSF10b gene, which encodes the TRAIL-R2 receptor, confers partial resistance to TRAIL-induced apoptosis. Specifically, TRAIL-R2 knockout (KO) cells exhibited reduced sensitivity to TRAIL across all tested concentrations, consistently showing higher viability compared to wild-type (WT) cells.

However, the TRAIL-R2 knockout cells were not completely resistant to TRAIL-induced apoptosis. This partial resistance can be attributed to the presence of TRAIL-R1 (DR4), another receptor for TRAIL. Since TRAIL can bind to both TRAIL-R1 and TRAIL-R2, the complete loss of TRAIL-induced apoptosis would require the knockout of both receptors. Therefore, while the absence of TRAIL-R2 reduces sensitivity, it does not eliminate the apoptotic response entirely due to the continued presence and functionality of TRAIL-R1.

Western blot analysis further supported these observations by demonstrating reduced levels of cleaved PARP, a marker of apoptosis, in TRAIL-R2 KO cells compared to WT cells treated with TRAIL. This indicates that TRAIL-R2 plays a critical role in mediating TRAIL-induced apoptosis in HeLa cells. Additionally, it was noted that TRAIL-R2 levels decreased upon TRAIL stimulation in WT cells. This reduction is due to the internalization and degradation of the TRAIL-R2 receptor following ligand binding, which reflects the receptor's role in the apoptosis signaling pathway.

These results underscore the pivotal role of TRAIL-R2 in TRAIL-induced apoptosis signaling in HeLa cells. Disruption of TRAIL-R2 leads to reduced sensitivity to TRAIL, highlighting the potential of targeting TRAIL-R2 in cancer therapies that aim to modulate TRAIL-induced apoptosis. Future studies should focus on elucidating the downstream mechanisms involved in TRAIL-R2-mediated apoptosis to enhance the understanding of this pathway and its implications in cancer biology.

3.6 References

- F. Jiang, J. A. Doudna, 2016, CRISPR–Cas9 Structures and Mechanisms, Annual Review of Biophysics
- M. Redman et al., 2016, What is CRISPR/Cas9?, BMJ journals
- S. Basu et al., 2010, Purification of Specific Cell Population by Fluorescence Activated Cell Sorting (FACS), JoVE Journal
- J.W. Tung et al., 2004, New approaches to fluorescence compensation and visualization of FACS data, ScienceDirect
- S.C. Taylor, A. Posch, 2014, The Design of a Quantitative Western Blot Experiment, BioMed Research International

4 Cell Death Modalities (Flow Cytometry)

4.1 Introduction

Cellular stress can lead to programmed cell death via apoptosis or necroptosis. Apoptosis is a caspase-dependent process, whereas necroptosis is a kinase-dependent pathway involving RIP1, RIP3, and MLKL. Both pathways can be distinguished morphologically or by blocking key proteins. HeLa and HT-29 cells serve as model systems for studying these pathways. HeLa cells, known for their type II extrinsic apoptosis signaling, express RIP1 but have low RIP3 levels, making them necroptosis-resistant. HT-29 cells express all proteins involved in both pathways and are used as a model for necroptosis.

4.2 Objective

The objective of this experiment is to distinguish between apoptosis and necroptosis in HeLa and HT-29 cells under various treatment conditions. We aim to induce apoptosis or necroptosis using TNF- α in combination with Birinapant, zVAD, and necrostatin-1, and then to identify the resulting cell death pathways by using Annexin V-eGFP and propidium iodide (PI) staining.

4.3 Materials and Methods

Materials:

- HeLa cells
- HT-29 cells
- TNF- α (20 μ g/ml stock concentration)
- Birinapant (1 mM stock concentration)
- zVAD-fmk (25 mM stock concentration)
- Necrostatin-1 (25 mM stock concentration)
- Annexin V-eGFP
- Propidium iodide (PI)
- 96-well plates (F-bottom)
- PBS
- Trypsin/EDTA
- Binding buffer
- Flow cytometer

Methods:

The experiment was performed according to the laboratory script. Detailed information on the procedure can be found in the laboratory script for this experiment.

4.4 Results

4.4.1 Seed Cells

Based on the cell count result of 32 cells per 10 μ l, we calculated the cell concentration as follows:

$$\text{Counted cells} = 32 \times 2 \times 10^4 = 640000 \text{ cells/ml}$$

we prepared medium For 30 wells (12 extra), the total volume of medium required is:

$$30 \text{ wells} \times 200\mu\text{l} = 6 \text{ ml}$$

Total Cells Needed:

$$30 \text{ wells} \times 30000 \text{ cells/well} = 900000 \text{ cells}$$

Volume of Cell Suspension Needed:

$$\frac{900000 \text{ cells}}{640000 \text{ cells/ml}} \approx 1.40 \text{ ml}$$

Total Volume to Prepare:

$$6 \text{ ml}(\text{total volume}) - 1.40 \text{ ml}(\text{cellsuspension}) = 4.6 \text{ ml}(\text{medium})$$

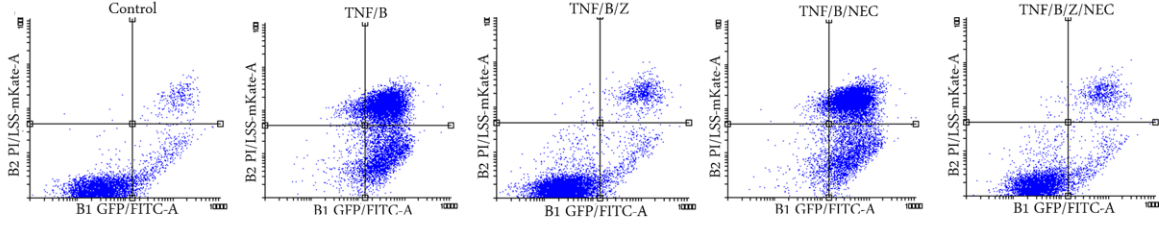
4.4.2 Treatment

HT-29 and HeLa cells are treated with various conditions to induce cell death pathways. Four different treatments are prepared: Control, TNF- α with Birinapant (TNF/B), TNF- α with Birinapant and zVAD (TNF/B z), and TNF- α with Birinapant and necrostatin-1 (TNF/B nec). Each treatment is prepared in a double-concentrated mix and added sequentially to the wells. The cells are incubated with the first mix for 30 minutes, followed by the addition of the second mix, and then incubated at 37°C for 24 hours. This setup allows the study of apoptosis, necroptosis, and NF-kB signaling pathways.

4.4.3 Flow Cytometry Analysis

The flow cytometry analysis of HeLa and HT-29 cells treated under different conditions was performed to assess the induction of apoptosis and necroptosis. The cells were stained with Annexin V-eGFP and propidium iodide (PI) to distinguish between live, apoptotic, and necrotic cells. The data was analyzed using Flowing software and visualized with dot plots.

HeLa Cell Lines



HT-29 Cell Lines

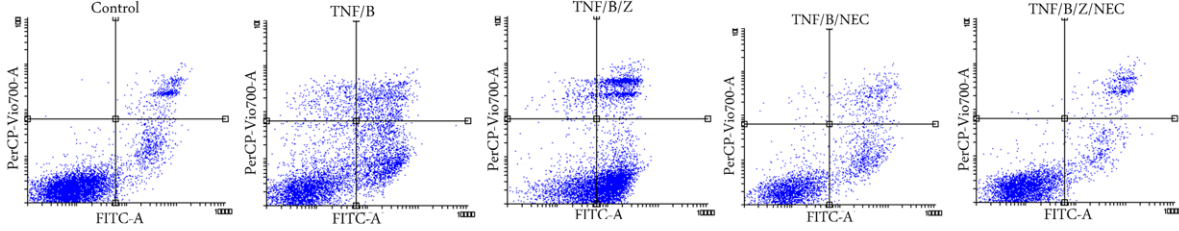


Figure 6: The flow cytometry analysis of HeLa and HT-29 cell lines was performed under various treatment conditions after a 24-hour incubation at 37°C. The cells were stained with Annexin V-eGFP (detected in the FITC-A channel) to identify apoptotic cells, and with propidium iodide (detected in the PerCP-Vio700-A channel for HT-29 cells and the SSC-mK-eT670-A channel for HeLa cells) to distinguish necrotic cells. Each dot plot represents one treatment condition, displaying the distribution of cells across different states—live, apoptotic, and necrotic—based on their respective staining profiles.

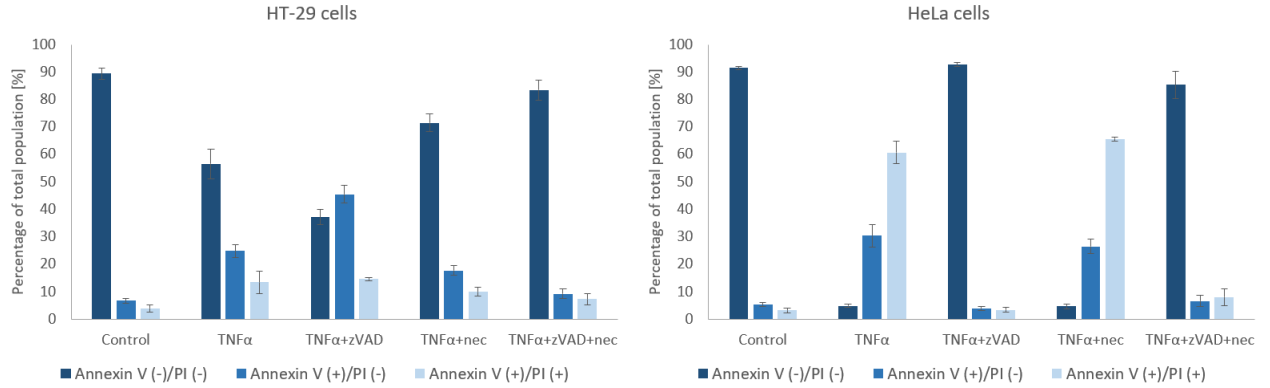


Figure 7: Bar charts of the flow cytometric analysis of HeLa and HT-29 cells. The error bars represented percentage error bars calculated as $\left(\frac{|Mean-X|}{X} \times 100\right)$. The corresponding data are available [here](#).

HeLa Cells (right-hand side of figure 7)

1. Control (Ctrl): The majority of HeLa cells were alive, with low levels of positivity for both Annexin V and PI staining, indicating minimal apoptosis or necrosis.

2. TNF- α /Birinapant (TNF/B): An increase in Annexin V-positive cells was observed, indicating a higher rate of apoptosis compared to the control. The number of PI-positive cells also increased, suggesting some level of necrosis or late apoptosis. Birinapant, an IAP antagonist, was used to enhance the apoptotic response induced by TNF- α .

3. TNF- α /Birinapant + zVAD (TNF/B z): The presence of zVAD, a caspase inhibitor, led to a reduction in Annexin V-positive cells compared to the TNF α treatment alone. This indicates that blocking caspase activity inhibits apoptosis. However, there was an increase in PI-positive cells, suggesting that necroptosis might be occurring.

4. TNF- α /Birinapant + Necrostatin-1 (TNF/B nec): The addition of necrostatin-1, an inhibitor of RIP1 kinase, reduced the number of PI-positive cells, indicating that necroptosis was effectively blocked. The level of apoptosis was similar to that observed with TNF α treatment alone.

5. TNF- α /Birinapant + zVAD + Necrostatin-1 (TNF/B z nec): This combination further reduced both Annexin V and PI-positive cells, suggesting that both apoptosis and necroptosis were effectively inhibited.

HT-29 Cells (left-hand side of figure 7)

1. Control (Ctrl): Similar to HeLa cells, the majority of HT-29 cells in the control group were alive, with low levels of apoptosis and necrosis.

2. TNF- α /Birinapant (TNF/B): A significant increase in Annexin V-positive cells was observed, indicating a higher rate of apoptosis. The increase in PI-positive cells was also more pronounced compared to HeLa cells, suggesting a higher susceptibility to necrosis or late apoptosis.

3. TNF- α /Birinapant + zVAD (TNF/B z): The inhibition of caspases with zVAD resulted in a marked decrease in Annexin V-positive cells, indicating reduced apoptosis. However, there was a substantial increase in PI-positive cells, suggesting enhanced necroptosis.

4. TNF- α /Birinapant + Necrostatin-1 (TNF/B nec): The addition of necrostatin-1 significantly reduced PI-positive cells, indicating effective inhibition of necroptosis. The level of apoptosis remained similar to the TNF α treatment.

5. TNF- α /Birinapant + zVAD + Necrostatin-1 (TNF/B z nec): This combination treatment resulted in a significant reduction in both Annexin V and PI-positive cells, indicating effective inhibition of both apoptosis and necroptosis.

Comparison of HeLa and HT-29 Cells

- **Apoptosis:** Both cell lines showed increased apoptosis upon TNF α treatment, but HT-29 cells were more sensitive to apoptotic induction compared to HeLa cells.

- **Necroptosis:** HeLa cells do not undergo necroptosis. The observed PI-positive staining in HeLa cells can be attributed to the loss of membrane integrity in apoptotic cells. In contrast,

HT-29 cells demonstrated a higher level of necroptosis, as evidenced by a significant increase in PI-positive cells when caspase activity was inhibited. This suggests that HT-29 cells are capable of necroptosis, which is evident when caspase inhibitors like zVAD are present.

- **Inhibition of Necroptosis:** The addition of necrostatin-1, an inhibitor of RIP1 kinase, did not affect the level of cell death in HeLa cells, as they do not undergo necroptosis. In HT-29 cells, necrostatin-1 effectively reduced necroptosis, but this effect was more pronounced when combined with zVAD, highlighting the dependency of HT-29 cells on caspase inhibition for necroptotic cell death. The lack of effect in HeLa cells indicates that necrostatin-1 does not impact apoptotic pathways in these cells.

Overall, these results demonstrate the distinct responses of HeLa and HT-29 cells to apoptosis and necroptosis induction and highlight the effectiveness of specific inhibitors in modulating these cell death pathways.

4.5 Discussion

In this experiment, we investigated the differential induction of apoptosis and necroptosis in HeLa and HT-29 cells under various treatment conditions. Flow cytometry analysis using Annexin V-eGFP and propidium iodide (PI) staining allowed us to distinguish between live, apoptotic, and necrotic cells. Our results highlight the distinct cellular responses to TNF- α and Birinapant, as well as the modulatory effects of zVAD and necrostatin-1.

HeLa Cells

HeLa cells are resistant to necroptosis due to low levels of RIP3. As a result, these cells primarily underwent apoptosis upon TNF- α /Birinapant treatment. The presence of zVAD effectively reduced apoptosis, but contrary to previous assumptions, it did not induce necroptosis. Instead, the data showed that zVAD completely prevented cell death, demonstrating that HeLa cells cannot switch to necroptosis because they lack the necessary components (such as RIP3) to execute this pathway. The addition of necrostatin-1 did not further affect cell death, confirming that HeLa cells are unable to undergo necroptosis.

HT-29 Cells

HT-29 cells, which express all key proteins involved in both apoptosis and necroptosis, exhibited a robust apoptotic response to TNF- α /Birinapant treatment. The combination with zVAD significantly increased necrotic cell death, indicating a switch to necroptosis. The addition of necrostatin-1 effectively inhibited this necroptotic response, reducing the number of PI-positive cells.

Comparison Between HeLa and HT-29 Cells

Both cell lines showed increased apoptosis in response to TNF- α /Birinapant treatment, with HT-29 cells being more sensitive to apoptosis than HeLa cells. This increased sensitivity in HT-29 cells is likely due to their expression of RIP3, which facilitates both apoptosis and necroptosis pathways. The presence of zVAD did not induce necroptosis in HeLa cells, aligning with the lack of RIP3 expression and indicating that HeLa cells are not capable of necroptosis. In HT-29 cells, however, zVAD induced necroptosis, and the addition of necrostatin-1 effectively blocked this response, underscoring the role of RIP1 kinase in necroptosis.

Overall, our results demonstrate the differential sensitivity of HeLa and HT-29 cells to apoptosis and necroptosis and highlight the importance of specific inhibitors in modulating these cell death pathways. The distinct responses observed in the two cell lines underline the complexity of cell death regulation and the need for targeted therapeutic strategies.

4.6 Conclusion

This study successfully distinguished between apoptosis and necroptosis in HeLa and HT-29 cells under various treatment conditions using flow cytometry. Our findings demonstrate:

- Apoptosis Induction: Both HeLa and HT-29 cells exhibited increased apoptosis upon treatment with TNF- α and Birinapant, with HT-29 cells showing a higher sensitivity.
- Necroptosis Induction: The presence of zVAD induced necroptosis in both cell lines, with a more significant effect observed in HT-29 cells.
- Inhibition of Necroptosis: Necrostatin-1 effectively inhibited necroptosis in HT-29 cells, reducing the number of necrotic cells. However, this inhibitor had no effect on HeLa cells, which are incapable of necroptosis.

These results underscore the potential of targeting specific cell death pathways for therapeutic intervention and provide valuable insights into the differential regulation of apoptosis and necroptosis in cancer cells.

4.7 References

- A.L. Givan, 2010, Flow Cytometry: An Introduction, Methods in Molecular Biology
A. Adan et al., 2016, Flow cytometry: basic principles and applications, Critical Reviews in Biotechnology
K.M. McKinnon, 2018 Flow Cytometry: An Overview, Current Protocols in Immunology

5 Time-Lapse Imaging

5.1 Introduction

The fluorescence ubiquitination cell cycle indicator (FUCCI2) is a two-color fluorescent protein-based system that utilizes red and green fluorescent proteins fused to cell cycle regulators, mCherry-cdt1 (amino acids 30 – 120) and mVenus-geminin (amino acids 1 – 110), to illustrate cell division within a cell population. This genetically encoded system is stably expressed in cell lines created for cell cycle experiments, such as those conducted by Murakami and Aida (PlosONE 2014). FUCCI2 allows for real-time observation of cell cycle phases, with cells in the G1 phase appearing colorless and those in the second half of the cell cycle appearing green. This system provides a powerful tool for studying cell cycle dynamics and the effects of cellular stress induced by starvation or drugs like nocodazole and aphidicolin, which can halt the cell cycle at different stages.

Nocodazole, an antineoplastic agent, interferes with microtubule polymerization, preventing the formation of the mitotic spindle and causing cells to arrest in mitosis, which results in a green, rounded appearance. Aphidicolin, a reversible inhibitor of DNA polymerase A and D, arrests cells at the beginning of the S phase, resulting in a light green appearance.

Serum starvation is another treatment used in cell cycle studies. By depriving cells of serum, which contains essential growth factors and nutrients, serum starvation induces a G1 phase arrest. This treatment is used to synchronize cell populations and study the effects of nutrient deprivation on cell cycle progression. Cells in a serum-starved state typically appear colorless in FUCCI2 imaging, reflecting their arrest in the G1 phase.

By using FUCCI2-expressing non-small-cell lung cancer cell lines, this experiment aims to visualize and quantify the effects of nocodazole, aphidicolin, and serum starvation on the cell cycle, providing insights into how these treatments influence cell cycle dynamics and cellular stress responses.

5.2 Objective

The objective of this experiment is to utilize the FUCCI2 system to observe and analyze cell cycle progression and arrest in NCI-H460 cells stably expressing mVenus-geminin. Specifically, we aim to:

1. Observe the effects of serum starvation, nocodazole, and aphidicolin on cell cycle progression.
2. Quantify the proportion of cells in different cell cycle phases (G1, S/G2, M, and dead cells) over time using epi-fluorescence wide field microscopy.
3. Analyze the duration of the G1, S/G2, and M phases in individual cells under each treatment condition.

5.3 Materials and Methods

Materials:

- NCI-H460 cells stably expressing mVenus-geminin
- RPMI medium without phenol red
- 10% FBS (Fetal Bovine Serum)
- Penicillin-Streptomycin (PenStrep)
- PBS (Phosphate Buffered Saline)
- Nocodazole (stock I: 5 mg/ml; stock II: 50 µg/ml)
- Aphidicolin (stock I: 5 mg/ml; stock II: 0.5 mg/ml)
- Four-chambered glass-bottom dish
- 1.5 ml Eppendorf cups
- Epi-fluorescence wide field microscope with climate chamber

Methods:

The experiment was performed according to the laboratory script. Detailed information on the procedure can be found in the laboratory script for this experiment.

5.4 Results

5.4.1 Seed Cells

Based on the cell count result of ≈ 26 cells per $10\ \mu l$, we calculated the cell concentration as follows:

$$\text{Counted cells} = 26 \times 2 \times 10^4 = 520000\ \text{cells/ml}$$

We prepared a suspension for five chambers (one chamber extra!), which means a total of $35000 \times 5 = 175000$ cells in $2.5\ ml$ suspension. To calculate the volume of cell culture needed, we performed the following calculation:

$$\text{Cell Culture volume needed} = \frac{175000\ \text{cells}}{520000\ \text{cells/ml}} \approx 0.337\ ml$$

$$\text{Medium volume needed} = 2.5\ ml(\text{Suspension needed}) - 0.337\ ml = 2.16\ ml$$

5.4.2 Sample Preparation and Imaging

On the next day, we treated the seeded NCI-H460 cells with cell cycle-arresting drugs and prepared them for fluorescence microscopy. After incubating the treated cells for 1 hour, we transported them to a pre-heated microscope stage and set up the Cell Observer microscope for time-lapse imaging over 48 hours. The microscopy data are available [here](#).

5.4.3 Image Analysis

The time-lapse imaging reveals the effects of different treatments on the cell cycle progression of NCI-H460 cells expressing mVenus-geminin. The control group shows normal cell proliferation, with cells transitioning through different phases of the cell cycle as indicated by the changes in GFP fluorescence over 48 hours. Serum starvation significantly slows down cell division, evident from the reduced number of cells and weaker GFP signals over time. Nocodazole treatment arrests cells in mitosis, leading to a high number of green-fluorescent, rounded cells at both 24 and 48 hours. Aphidicolin treatment causes cells to accumulate in the S phase, resulting in a progressive increase in green fluorescence. These observations confirm the expected outcomes of the treatments, with nocodazole blocking cells in mitosis and aphidicolin inhibiting DNA replication, thereby arresting the cells in the S phase.

The bar diagram further quantifies these observations by showing the proportion of cells in various cell cycle phases under each treatment condition. The control group exhibits a balanced distribution across the cell cycle phases. Serum starvation results in an increased proportion of cells in the G0/G1 phase, indicating cell cycle arrest. Nocodazole treatment leads to a marked increase in the proportion of mitotic cells, corroborating the microscopy data of rounded, green cells. Aphidicolin treatment shows a substantial accumulation of cells in the S phase, aligning with the progressive increase in green fluorescence observed in the images. Overall, the combined analysis of the time-lapse images and the bar diagram provides a comprehensive understanding of how these treatments affect cell cycle progression.

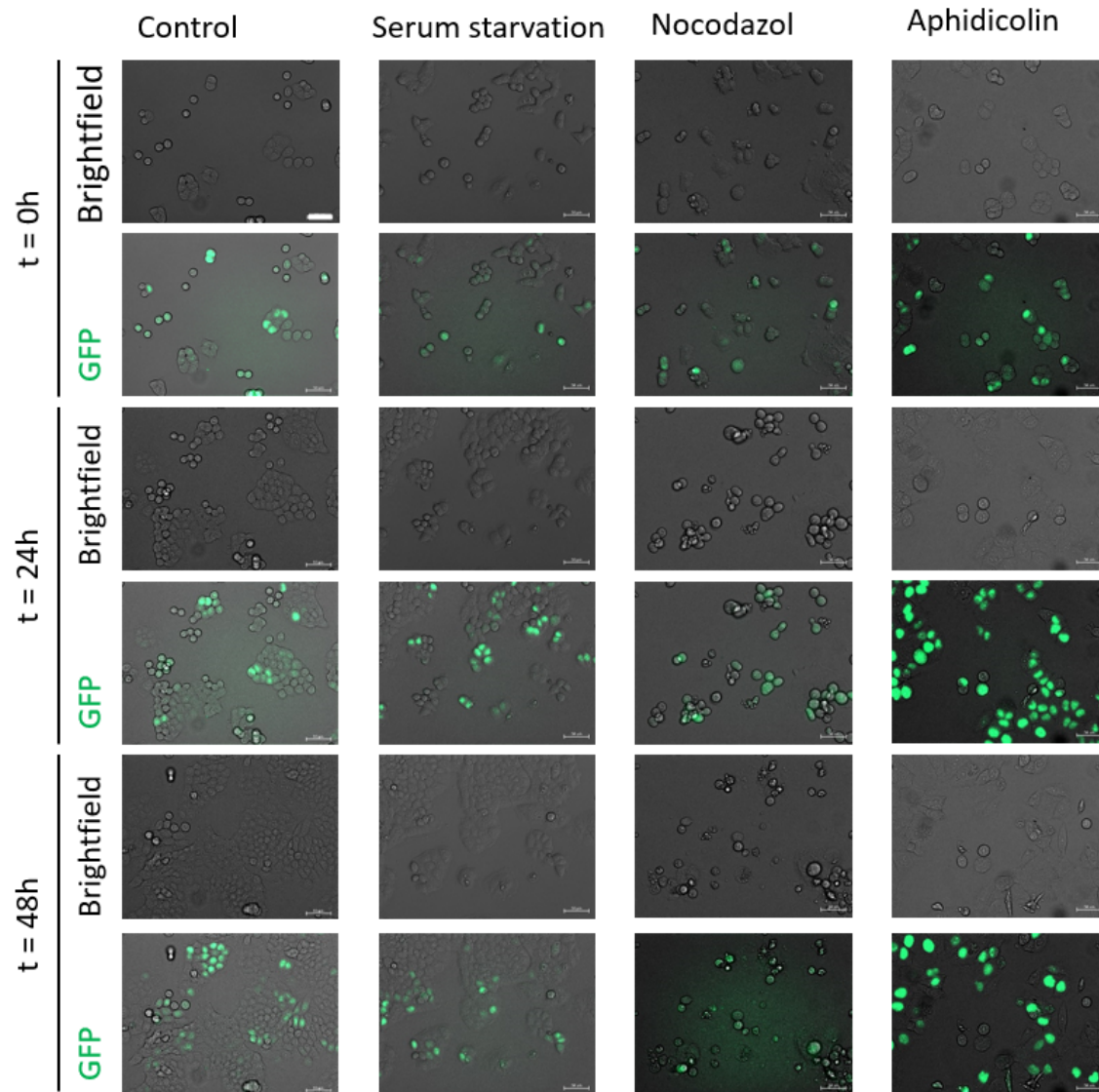


Figure 8: This set of images shows the time-lapse imaging of NCI-H460 cells under different treatment conditions: control, serum starvation, nocodazole, and aphidicolin at various time points (0h, 24h, 48h). The first row displays brightfield images at time 0h for each treatment condition, followed by GFP images for the same time point. Subsequent rows show brightfield and GFP images at 24h and 48h. In the control group, cells show normal growth and division over time. Under serum starvation, there is a notable reduction in cell proliferation by 48h. Nocodazole treatment results in cells arrested in mitosis, appearing rounded and green, particularly evident at 24h and 48h. Aphidicolin treatment causes cells to arrest at the beginning of the S phase, leading to an accumulation of green-fluorescent cells over time.

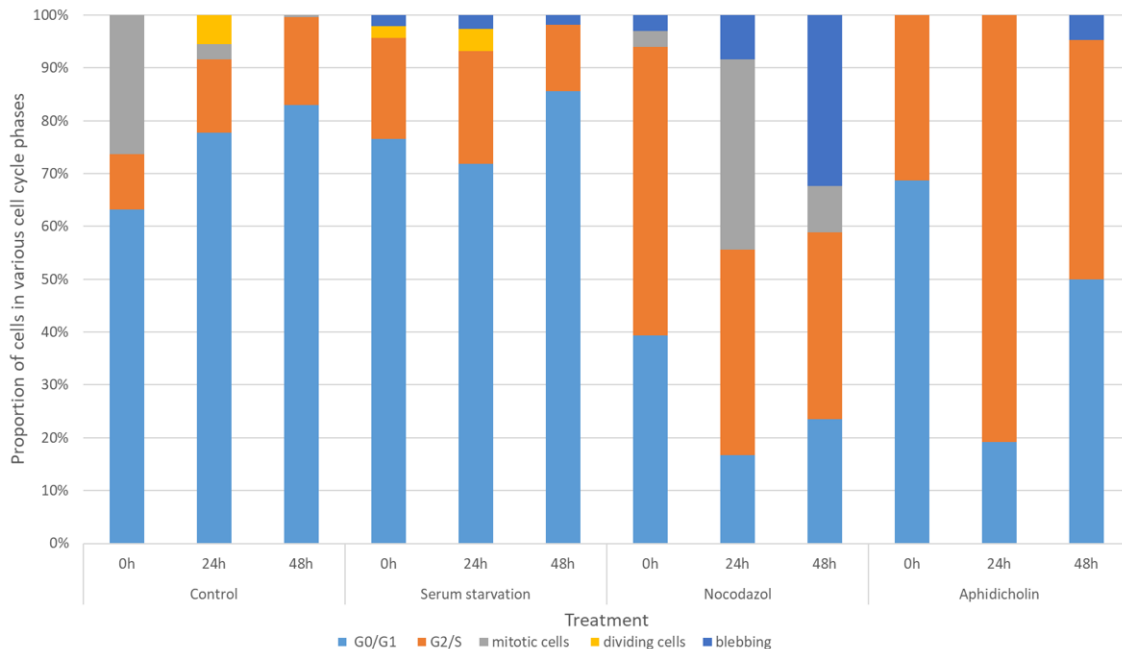


Figure 9: This bar chart illustrates the proportion of cells in various cell cycle phases under different treatment conditions over time. The phases are categorized as follows: G0/G1 phase (light blue), G2/S phase (orange), mitotic cells (gray), dividing cells (yellow), and blebbing cells (dark blue). The control group displays a balanced distribution across these phases. Serum starvation leads to an increased proportion of cells in the G0/G1 phase over time compared to other treatments (Nocodazole and Aphidicolin), although this increase is not significant relative to the control. Nocodazole significantly elevates the proportion of mitotic and dividing cells, while Aphidicolin causes an accumulation of cells in the G2/S phase. These results highlight the distinct impacts of each treatment on cell cycle progression. The complete dataset is available [here](#).

5.5 Discussion

The time-lapse imaging and quantification results demonstrate the distinct effects of serum starvation, nocodazole, and aphidicolin on cell cycle progression in NCI-H460 cells. Under control conditions, cells proliferate normally, passing through various phases of the cell cycle. Serum starvation results contrary to what might be anticipated, serum starvation does not lead to a significant increase in the proportion of cells in the G0/G1 phase compared to the control. This suggests that, although serum starvation is expected to induce cell cycle arrest at G0/G1 due to nutrient deprivation, the data does not show a marked increase in this phase, indicating that the effect of serum starvation on cell cycle progression may be less pronounced or inconsistent with expectations. Nocodazole, which disrupts microtubule polymerization, causes cells to arrest in mitosis, resulting in a high number of rounded, green-fluorescent cells. Aphidicolin, an inhibitor of DNA polymerase, arrests cells at the beginning of the S phase, leading to an accumulation of cells in the G2/S phase. These observations are consistent with the expected mechanisms of action of the respective treatments.

5.6 Conclusion

The FUCCI2 system effectively visualizes the impact of different treatments on cell cycle progression in NCI-H460 cells. Serum starvation arrests cells in the G0/G1 phase, nocodazole in mitosis, and aphidicolin at the beginning of the S phase. Time-lapse imaging, combined with quantitative analysis, provides a comprehensive view of cell cycle dynamics under various conditions, facilitating the understanding of cellular responses to stress and drug treatment.

5.7 References

R.D.Gallego et al., 2019, Time-lapse imaging: the state of the art, *Biology of Reproduction*
E. Meijering et al., 2008, Time-Lapse Imaging, *Microscope Image Processing*

6 Image Processing And Analysis (FIJI)

6.1 Introduction

The apoptosis process involves intricate cellular events, including the activation of effector caspases like Caspase-3, which cleave specific substrates such as DEVD FRET probes. Monitoring these events requires precise image processing techniques to accurately quantify changes in fluorescent signals over time.

6.2 Objective

The objective of this experiment is to trace individual cells undergoing apoptosis, analyze effector caspase activity through the cleavage of DEVD FRET probes, and visualize these dynamics using fluorescence microscopy and image analysis techniques.

6.3 Materials and Methods

The experiment was performed according to the laboratory script. Detailed information on the procedure can be found in the laboratory script for this experiment.

6.4 Results

In this experiment, we used ImageJ to analyze single-cell images obtained from fluorescence microscope time-lapse experiments, focusing on DEVD FRET-probe expressing cells undergoing apoptosis. The analysis aimed to trace an individual cell's apoptosis process and assess effector caspase activity.

The raw digital images captured CFP emission, YFP emission, and the FRET channel, containing background noise that was reduced or removed prior to quantitative analysis. Background subtraction was performed using a macro to obtain background-corrected stacks for the CFP, YFP, and FRET channels. Segmentation was achieved by thresholding the YFP channel, creating binary masks, and applying these masks to the CFP and FRET stacks.

A ratiometric CFP/FRET image stack was created by dividing the background-corrected CFP stack by the FRET stack and applying a scaling factor to visualize FRET substrate cleavage as changes in signal intensity. This ratiometric stack displayed changes in cellular intensity, indicating caspase-mediated cleavage events.

We measured the CFP/FRET intensity changes in individual cells over time. ROIs were drawn around cells undergoing apoptosis, and the average signal intensities were recorded and exported to MS Excel for further analysis. The data was plotted against time to visualize substrate cleavage kinetics.

The results are shown in Figures 10 and 11.

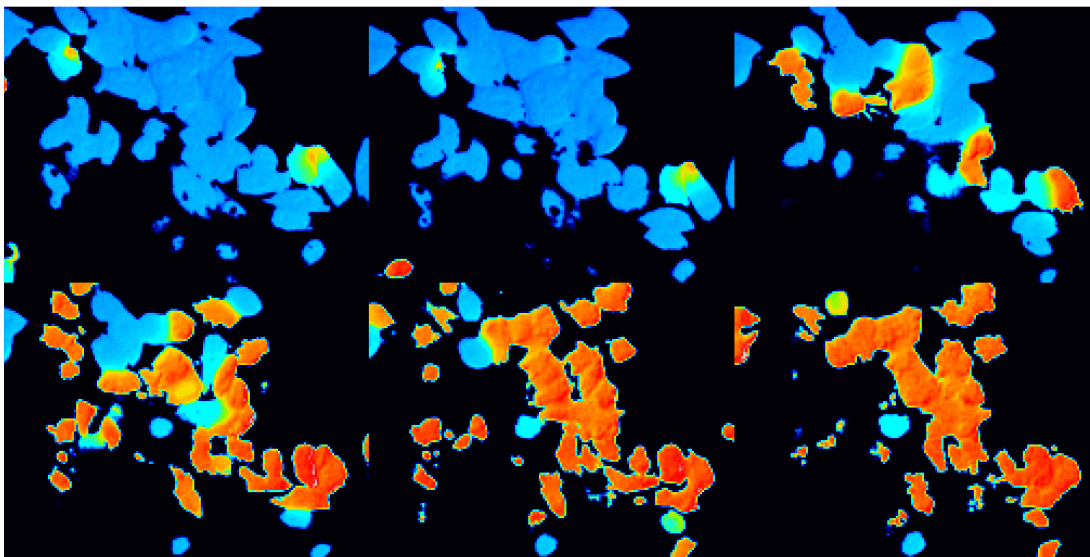


Figure 10: Representative images of a cell undergoing apoptosis at different time points (frames 1, 60, 120, 180, 240, 300)

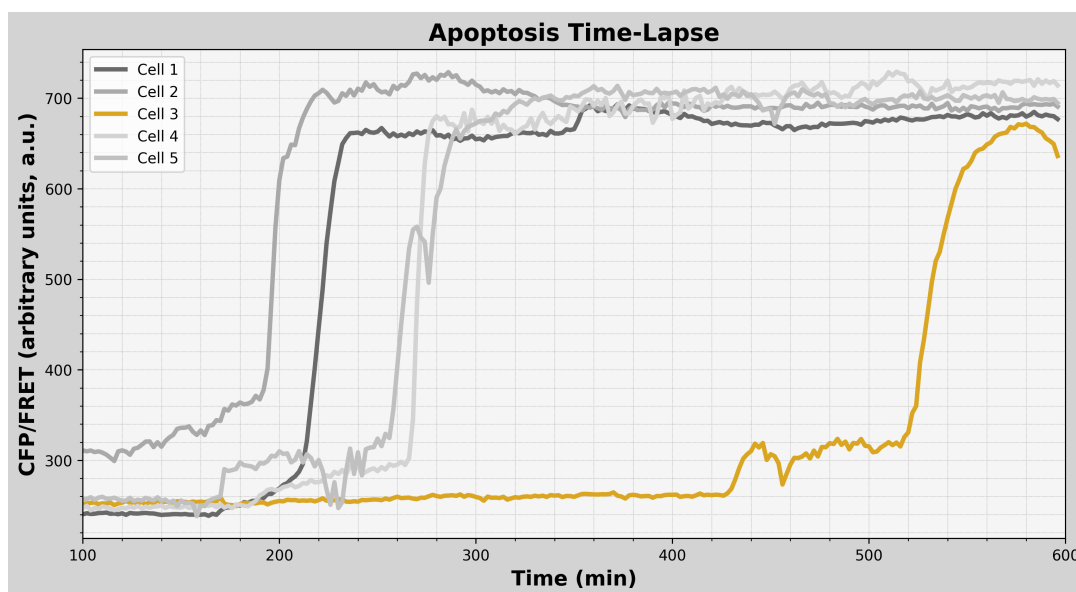


Figure 11: Five different cells which underwent apoptosis in different time points. The corresponding data is available [here](#).

6.5 Discussion

The analysis of DEVD FRET-probe expressing cells provided insights into the dynamics of caspase activity during apoptosis. The background subtraction and cell segmentation steps were crucial in obtaining accurate measurements of CFP and FRET emissions. By creating a

rationetric CFP/FRET stack, we could visualize the cleavage of the FRET probe, indicative of caspase activity, as a change in cellular intensity. This method allowed for the precise tracking of apoptosis in individual cells.

The resulting intensity changes were consistent with the expected increase in CFP and decrease in FRET emissions upon caspase activation. The use of a scaling factor during rationetric image creation is necessary to stretch the results across the bit depth of the resulting image, ensuring that intensity changes is adequately visualized.

One limitation of this approach is the potential impact of refocusing the microscope during image acquisition, which could affect the intensity traces. Additionally, the static ROIs drawn around cells may not fully account for cell movement, potentially influencing the accuracy of the intensity measurements.

6.6 Conclusion

This experiment successfully demonstrated the use of ImageJ for analyzing apoptosis in DEVD FRET-probe expressing cells. The background subtraction, cell segmentation, and rationetric image creation techniques allowed for accurate tracking of caspase activity over time. The results confirmed the expected changes in CFP and FRET emissions upon caspase activation, providing valuable insights into the apoptosis process at the single-cell level.

6.7 References

T.J. Collins, 2007, ImageJ for Microscopy, BioTechniques
M.D. Abràmoff et al.,2004, Image Processing with ImageJ; BioPhotonics

7 Confocal Microscopy

7.1 Introduction

Confocal microscopy, also known as laser scanning microscopy (LSM), is a powerful technique for creating high-resolution images of living cells. This technique allows for the visualization of subcellular structures and proteins tagged with fluorescent markers. Unlike electron microscopy and atomic force microscopy, confocal microscopy can be used with living cells, making it invaluable for studying dynamic biological processes in real-time. In this experiment, we use confocal microscopy to observe the translocation of Cytochrome c, tagged with Green Fluorescent Protein (GFP), from mitochondria to the cytosol in HeLa cells treated with an apoptosis-inducing drug, TRAIL (TNF-related apoptosis-inducing ligand).

7.2 Objective

The objective of this experiment is to visualize and monitor the translocation of Cytochrome c from the mitochondria to the cytosol in HeLa cells in response to TRAIL-induced apoptosis. This will be achieved by staining the cells with specific dyes and using confocal microscopy to capture images at different time points.

7.3 Materials and Methods

Materials:

- HeLa cells stably expressing Cytochrome c-GFP
- RPMI medium with 10% Fetal Bovine Serum (FBS) and Penicillin-Streptomycin (Pen-Strep)
- Four-chambered glass-bottom dish
- Apoptosis-inducing drug TRAIL
- Pan-caspase inhibitor Z-VAD-FMK (zVAD)
- Tetramethylrhodamine methyl ester (TMRM)
- MitoTracker deep red
- HOECHST 33342
- Sterile Eppendorf cups
- Incubator
- Confocal microscope

Methods:

The experiment was performed according to the laboratory script. Detailed information on the procedure can be found in the laboratory script for this experiment.

7.4 Results

7.4.1 Seed Cells

Based on the cell count result of 27 cells per $10\ \mu\text{l}$, we calculated the cell concentration as follows:

$$\text{Counted cells} = 27 \times 2 \times 10^4 = 540000\ \text{cells/ml}$$

We prepared a suspension for five chambers (one chamber extra!), which means a total of $30000 \times 5 = 150000$ cells in $2.5\ \text{ml}$ suspension. To calculate the volume of cell culture needed, we performed the following calculation:

$$\text{Cell Culture volume needed} = \frac{150000\ \text{cells}}{540000\ \text{cells/ml}} = 0.278\ \text{ml}$$

$$\text{Medium volume needed} = 2.5\ \text{ml}(\text{Suspension needed}) - 0.278\ \text{ml} = 2.222\ \text{ml}$$

7.4.2 Cell Treatment

HeLa cells were subjected to the following treatments:

- **Control (Chamber 1):** Cells were untreated, serving as a baseline.
- **TRAIL (Chamber 2):** Cells were exposed to TRAIL to induce apoptosis.
- **TRAIL + MitoTracker deep red (Chamber 3):** Cells were treated with TRAIL and stained with MitoTracker deep red for enhanced mitochondrial visualization.
- **TRAIL + zVAD (Chamber 4):** Cells were treated with TRAIL in the presence of the pan-caspase inhibitor zVAD to block apoptosis.

7.4.3 Imaging and Observations

Confocal microscopy imaging of HeLa cells stably expressing Cytochrome c-GFP was performed under various treatment conditions, as shown in figure 12.

The analysis focused on visualizing Cytochrome c translocation from mitochondria to the cytosol in response to TRAIL-induced apoptosis, with the following key observations:

- **Control (Condition 1):** In the absence of TRAIL treatment, Cytochrome c-GFP was localized within mitochondria, as evidenced by the overlap with TMRM staining, which marks active mitochondria (red). The nuclei were stained with HOECHST (blue), showing normal cellular morphology without signs of apoptosis.

- **TRAIL Treatment (Condition 2):** Upon TRAIL treatment, there was (should be !) a noticeable release of Cytochrome c from mitochondria into the cytosol. This is observed as a decrease in the overlap between Cytochrome c-GFP and TMRM staining, with an increase in the diffuse green fluorescence (Cytochrome c-GFP) throughout the cytosol. The blue HOECHST staining indicates that nuclei remain intact, but the overall distribution of Cytochrome c has shifted from mitochondria to the cytosol.

- **TRAIL and MitoTracker deep red (Condition 3):** TRAIL treatment led to Cytochrome c release from mitochondria, and the use of MitoTracker Deep Red enhanced mitochondrial visualization. MitoTracker Deep Red covalently binds to mitochondria, maintaining visibility even when the mitochondrial membrane potential is lost. In contrast, TMRM staining, which accumulates in mitochondria with intact membrane potential, would have shown a loss of staining as mitochondria lose their membrane potential during apoptosis. The MitoTracker Deep Red results demonstrate the complete dissociation of Cytochrome c from mitochondria, indicating effective mitochondrial outer membrane permeabilization (MOMP) while highlighting the differences in staining methods.

- **TRAIL and zVAD (Condition 4):** The addition of the pan-caspase inhibitor zVAD effectively inhibited TRAIL-induced apoptosis, as shown by the reduced translocation of Cytochrome c-GFP to the cytosol. The presence of zVAD resulted in similar fluorescence patterns to the control, with Cytochrome c remaining largely within mitochondria. This suggests that zVAD successfully blocked the apoptosis pathway, preventing Cytochrome c release.

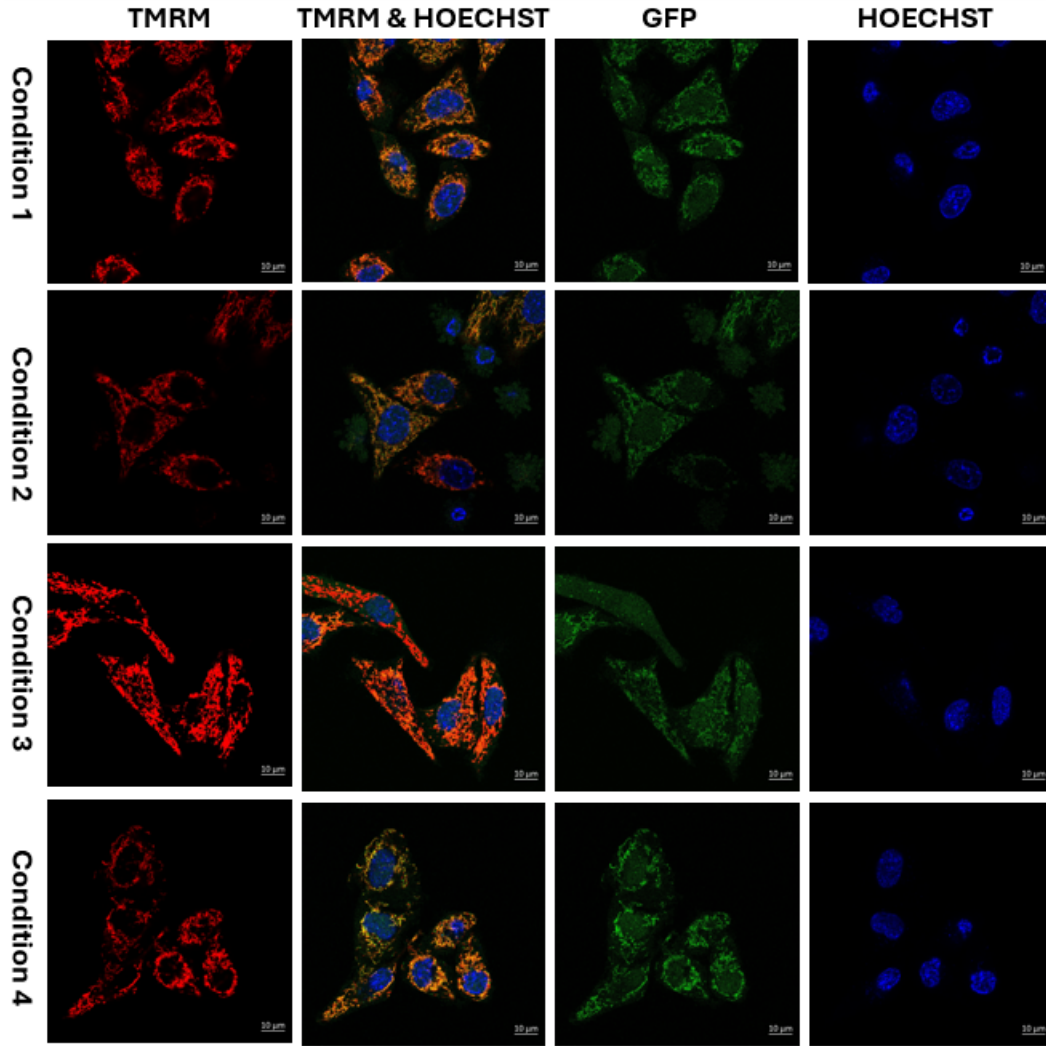


Figure 12: Confocal microscopy analysis of HeLa cells treated with TRAIL and various dyes. The 4×4 image matrix presents confocal microscopy images of HeLa cells stably expressing Cytochrome c-GFP under different conditions. Rows represent experimental treatments: condition 1 (Control, TMRM and HOECHST), condition 2 (TRAIL, TMRM and HOECHST), condition 3 (TRAIL, MitoTracker deep red and HOECHST), and condition 4 (TRAIL, zVAD, TMRM and HOECHST). Columns represent staining channels: TMRM (red, active mitochondria), TMRM and HOECHST (red and blue, mitochondria and nuclei), GFP (green, Cytochrome c-GFP translocation), and HOECHST (blue, nuclei). Images show TRAIL-induced mitochondrial outer membrane permeabilization and Cytochrome c release in conditions 2 and 3, with zVAD inhibiting this effect in condition 4. The combined fluorescence highlights the spatial relationships between mitochondria, Cytochrome c distribution, and nuclei under each treatment condition.

7.5 Discussion

The results from confocal microscopy provide valuable insights into the dynamics of Cytochrome c release during TRAIL-induced apoptosis. The observed translocation of Cytochrome c-GFP from mitochondria to the cytosol upon TRAIL treatment aligns with the

expected apoptotic response, where Cytochrome c acts as a key pro-apoptotic signal. The use of TMRM and MitoTracker deep red enabled effective visualization of mitochondrial integrity and Cytochrome c distribution, confirming that TRAIL-induced apoptosis involves MOMP and subsequent Cytochrome c release.

The inhibition of Cytochrome c translocation by zVAD underscores the role of caspases in the apoptosis pathway. By blocking caspase activation, zVAD effectively prevents the apoptotic cascade, illustrating the crucial role of caspases in regulating mitochondrial outer membrane permeabilization and Cytochrome c release.

The differential staining channels and experimental conditions in the imaging setup allowed for a comprehensive assessment of mitochondrial and cytosolic fluorescence, revealing the impact of TRAIL treatment on mitochondrial integrity and Cytochrome c distribution. This detailed analysis provides a clear depiction of the cellular response to apoptosis-inducing stimuli and the potential for pharmacological intervention.

7.6 References

- A. Nwaneshiudu et al., 2012, Introduction to confocal microscopy, Journal of Investigative Dermatology
- A.D. Elliott, 2019, Confocal Microscopy: Principles and Modern Practices, Current Protocols in Cytometry

8 Cell Viability Assay (MTT)

8.1 Introduction

The cell viability assay is a crucial experimental procedure for assessing the effects of various treatments on cell survival and proliferation. This experiment focuses on evaluating the dose-response relationship of HeLa cells treated with TRAIL and cycloheximide (CHX), as well as with other compounds chosen by students. The MTT assay, which measures cell viability based on the metabolic activity of living cells, is employed to quantify the effects of these treatments. By understanding how different substances influence cell viability, we can gain insights into their potential therapeutic or cytotoxic effects.

8.2 Objective

The objective of this experiment is to determine the impact of TRAIL and CHX, as well as other student-selected compounds, on the viability of HeLa cells. This will be achieved through a dose-response analysis using the MTT assay to measure cell viability after treatment. The results will help elucidate the potential synergistic, additive, or antagonistic effects of the combined treatments.

8.3 Materials and Methods

Materials:

- HeLa cells stably expressing Cytochrome-c-GFP
- RPMI medium with 10% FBS and PenStrep
- 96-well plates
- Eppendorf cups
- Sterile water
- TMRM, MitoTracker deep red, and HOECHST 33342 dyes
- TRAIL stock solution (50 µg/ml)
- CHX stock solution (200 µg/ml)
- MTT solution (5 mg/ml)
- Methanol
- Tecan plate reader (TECAN SPARK)
- Pipettes and sterile tips
- Thermomixer
- Incubator at 37°C

Methods:

The experiment was performed according to the laboratory script. Detailed information on the procedure can be found in the laboratory script for this experiment. In summary, HeLa cells were seeded in 96-well plates and treated with various concentrations of TRAIL, CHX, and other compounds. After 24 hours of treatment, cell viability was assessed using the MTT assay, with absorbance measured at 570 nm to determine the viability percentage relative to control wells.

8.4 Results

The MTT assay was performed to assess the viability of HeLa cells after treatment with various concentrations of TRAIL and CHX (cycloheximide), as well as with different dilutions of a lemon-ginger water solution in combination with TRAIL. The results are depicted in Figure 13.

In the CHX treatment group, a dose-dependent decrease in cell viability was observed with increasing concentrations of CHX combined with TRAIL. At 0 $\mu\text{g/mL}$ CHX, cell viability was high across all TRAIL concentrations, showing only a slight reduction at the highest TRAIL concentration, where TRAIL alone had the most substantial effect. As the concentration of CHX increased, cell viability significantly decreased, indicating a synergistic effect between CHX and TRAIL. The combination of 1 $\mu\text{g/mL}$ CHX and 1 ng/mL TRAIL resulted in the lowest cell viability, suggesting that CHX effectively sensitizes HeLa cells to TRAIL-induced apoptosis.

For the Lemon-ginger treatment group, different dilutions were tested in combination with TRAIL. At a 1:1000 dilution of Lemon-ginger, cell viability remained relatively high, even with increasing TRAIL concentrations. At a 1:100 dilution, a notable decrease in cell viability was observed, particularly at higher TRAIL concentrations, indicating an additive or synergistic effect. The 1:10 dilution showed the most significant reduction in cell viability, especially at 1 ng/mL TRAIL, highlighting a potent interaction between the Lemon-ginger solution and TRAIL.

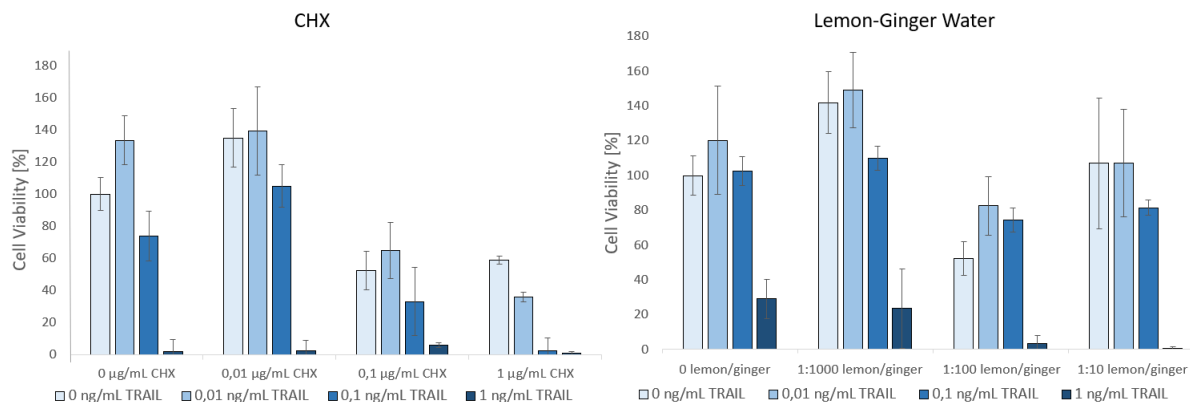


Figure 13: MTT Assay Results. The left panel displays the effect of CHX treatment on cell viability. Each group of four bars represents different CHX concentrations (0 to 1 µg/mL, from left to right). Within each CHX concentration, the bars correspond to increasing concentrations of TRAIL (0, 0.01, 0.1, and 1 ng/mL, from left to right). The right panel shows the effect of lemon-ginger water on cell viability. Each group of four bars represents a different concentration of lemon-ginger water (0, 1:1000, 1:100, and 1:10, from left to right), tested in combination with TRAIL at varying concentrations. The corresponding data is available [here](#).

8.5 Discussion

The results of this experiment demonstrate that both CHX and lemon-ginger solutions can enhance the cytotoxic effects of TRAIL on HeLa cells. CHX, a known inhibitor of protein synthesis, effectively sensitizes HeLa cells to TRAIL-induced apoptosis. This synergistic effect is likely due to CHX's ability to inhibit the synthesis of proteins that would otherwise inhibit apoptosis, thereby making the cells more susceptible to TRAIL's apoptotic signaling.

The lemon-ginger solution also showed a dose-dependent effect on cell viability when combined with TRAIL. At higher dilutions (1:1000), the solution had a minimal impact on cell viability, whereas more concentrated solutions (1:100 and 1:10) significantly enhanced the cytotoxic effects of TRAIL. This suggests that components within the lemon-ginger solution may have bioactive properties that can potentiate TRAIL's effectiveness. The exact mechanism of action remains unclear, but it could involve the modulation of apoptotic pathways or other cellular stress responses.

These findings indicate that natural compounds, like those found in lemon-ginger solutions, could be explored as potential adjuvants in TRAIL-based therapies. However, further studies are needed to identify the active components and understand their mechanisms of action.

8.6 Conclusion

In conclusion, the co-treatment of HeLa cells with CHX and TRAIL, as well as lemon-ginger solution and TRAIL, significantly reduces cell viability in a dose-dependent manner.

CHX sensitizes cells to TRAIL-induced apoptosis, demonstrating a clear synergistic effect. Similarly, lemon-ginger solutions at higher concentrations enhance the cytotoxic effects of TRAIL, suggesting potential bioactive properties. These results provide valuable insights into the potential use of combination therapies involving TRAIL and natural compounds for cancer treatment. Further research is warranted to elucidate the mechanisms underlying these interactions and to explore their clinical applications.

8.7 References

P. Kumar et al., 2018, Analysis of Cell Viability by the MTT Assay, Cold Spring Harb Protoc
A.V.B. Pintor et al., 2020, MTT versus other cell viability assays to evaluate the biocompatibility of root canal filling materials: a systematic review, International Endodontic Journal

9 Mathematical Modelling (ApoptoCoop)

9.1 Introduction

Apoptosis, or programmed cell death, is a fundamental biological process that involves a highly regulated sequence of events. The execution phase of apoptosis is initiated by the permeabilization of the outer mitochondrial membrane, leading to the release of key effector proteins such as cytochrome-c (cyt-c) and Smac into the cytosol. These proteins activate a cascade of events involving the formation of the apoptosome, activation of caspase-9 and caspase-3, and subsequent cleavage of numerous substrates. The intricate interplay of these proteins and reactions forms a complex, non-linear signalling network.

Mathematical modelling, specifically through systems of coupled ordinary differential equations (ODEs), allows for a comprehensive analysis of this network. By simulating the temporal development of reactant concentrations and their interactions, researchers can gain mechanistic insights that are challenging to obtain through experimentation alone. The ApoptoEverything software facilitates such simulations, providing a platform to model the apoptosis execution phase with high detail and to compare these simulations with experimental data. This approach is particularly useful for studying variations in apoptosis signalling across different cell types and for prognosticating patient outcomes in diseases such as colorectal cancer.

9.2 Objective

The objective of this experiment is to utilize systems modelling to simulate and understand the kinetics of the apoptosis execution phase in different cell lines. By employing the ApoptoEverything software, which implements a detailed ODE model of apoptosis, we aim to investigate the dynamics of key apoptosis proteins and their interactions. This will involve comparing experimental results with simulations, analyzing the effects of varying protein concentrations, and exploring the prognostic implications of apoptosis competency in colorectal cancer patients.

9.3 Materials and Methods

Materials:

- Software: ApoptoEverything (requires Matlab or Matlab Runtime Environment)
- HeLa cells
- HCT116 cells
- Procaspase-3 (CASP3)
- Procaspase-9 (CASP9)
- XIAP
- Smac (mitoSMAC)
- Apaf-1

- Cytochrome C (mitoCytC)
- Matlab version 9.9/2020b or Matlab runtime version 9.9

Methods:

The experiment was performed according to the laboratory script. Detailed information on the procedure can be found in the laboratory script for this experiment. In summary, the ApoptoEverything software was used to simulate the apoptosis execution phase by inputting measured protein concentrations and other parameters. Various simulations were conducted to observe the kinetics of apoptosis signalling, the effects of protein concentration variations, and the prognostic implications for colorectal cancer patients based on apoptosis competency.

9.4 Results

9.4.1 Experiment 1

The simulation results, figure 14 indicate that the C3 cleavage percent (Y-axis) increases faster in HeLa cells compared to HCT116 cells, resulting in a smoother curve for HCT116 cells.

Possible Reasons Based on Simulation Data:

- **Reactant Concentrations:** HCT116 cells have lower initial concentrations of key apoptotic proteins (e.g., CASP3 and CASP9) compared to HeLa cells. This lower concentration can lead to a slower activation of the apoptosis pathway, resulting in a smoother increase in C3 cleavage percent.

- **Inhibitor Effect:** The concentration of inhibitors like XIAP is relatively higher in HCT116 cells, which can delay the cleavage of caspase-3. XIAP binds to caspase-3 and caspase-9, inhibiting their activity and thus slowing down the apoptosis process.

- **Protein Interactions:** The dynamics of protein interactions might be different between the two cell lines. In HeLa cells, the interactions might be more rapid and efficient, leading to a quicker activation of caspase-3.

- **Cellular Environment:** The model does not account for differences in mitochondrial health or other cellular factors. Therefore, differences observed in the simulations are due to variations in protein concentrations and inhibitors alone.

Note: Proteins and their interaction dynamics are consistent across cell lines, so observed variations are due to differences in concentrations of proteins and inhibitors, not changes in protein interactions. Additionally, the model does not incorporate mitochondrial health or other environmental factors, so differences observed are solely based on protein concentrations and inhibitors.

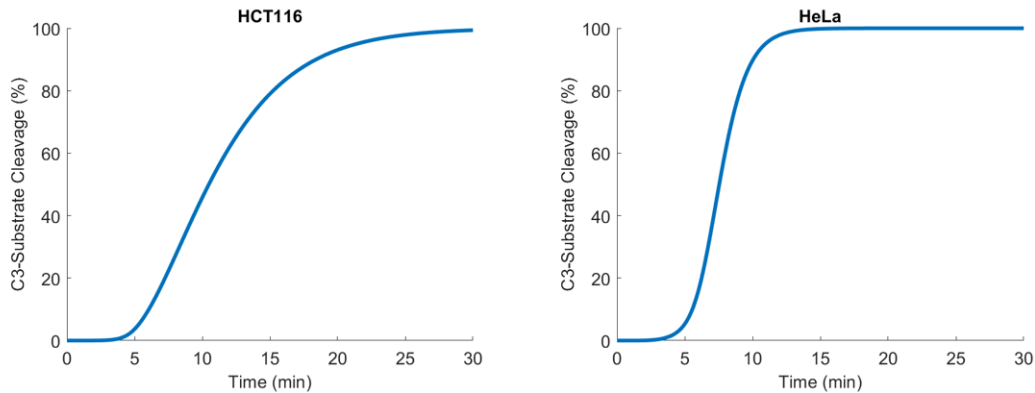


Figure 14: Comparison of C3 Cleavage Percent Over Time in HeLa and HCT116 Cells

9.4.2 Experiment 2

Subfigure HeLa 2:

The simulations for figure mode 2 show the initial and peak concentrations of CASP3, CASP9, XIAP, mitoSMAC, APAF1, and mitoCytC in HeLa cells. This mode helps in understanding the initial activation phase and the peak response of the apoptotic pathway.

Subfigure HeLa 3:

Figure mode 3 provides insights into the intermediate stages of apoptosis, highlighting how the concentrations of various reactants change over time. It shows the progression of apoptotic signaling and the formation of apoptotic complexes.

Subfigure HeLa 4:

The fourth mode illustrates the late stages of apoptosis, including the full activation of CASP3 and CASP9, and the downregulation of XIAP due to its inhibition by SMAC. It captures the decline in the levels of initiator and effector caspases as the apoptosis process progresses to completion.

Subfigure HeLa 5:

This mode focuses on the degradation and clearance of apoptotic components, showing how the concentrations of APAF1, mitoCytC, and other apoptotic proteins return to baseline levels, completing the apoptosis cycle.

Transient vs. Stable Complexes:

- **Stable Complexes:** Stable complexes, such as those involving XIAP binding to CASP9, show minimal changes over time. These complexes maintain a consistent presence, preventing premature activation of the apoptosis pathway.

- **Transient Complexes:** Transient complexes, such as the APAF1-initiator caspase complex, exhibit significant fluctuations. These complexes form rapidly upon apoptotic signal initiation and disassemble quickly once their function is complete, contributing to the dynamic nature of apoptosis regulation.

Note: Considering the different APAF1 oligomers, only the heptameric form is stable over time, while other forms are more transient. Stable complexes, such as those involving XIAP binding to CASP9, show minimal changes over time. In contrast, transient complexes, such as the APAF1-initiator caspase complex, exhibit significant fluctuations, contributing to the dynamic nature of apoptosis regulation.

Interpretation of Results Based on Apoptotic Proteins:

- **CASP3 and CASP9:** The rapid cleavage and activation of CASP3 and CASP9 in HeLa cells, as shown in figure mode 2 and 3, indicate an efficient and swift apoptotic response. This is in contrast to a more gradual response seen in HCT116 cells.

- **XIAP:** The levels of XIAP, an inhibitor of apoptosis, show how its presence can delay apoptosis by binding to and inhibiting caspases. The reduction in XIAP levels over time in figure mode 4 demonstrates the eventual overcoming of this inhibition by SMAC.

- **mitoSMAC and APAF1:** The release of mitoSMAC from the mitochondria and the subsequent formation of the apoptosome with APAF1 are critical steps in the amplification of the apoptotic signal. The kinetics of these events vary between cell types and are crucial for understanding the differential apoptotic responses.

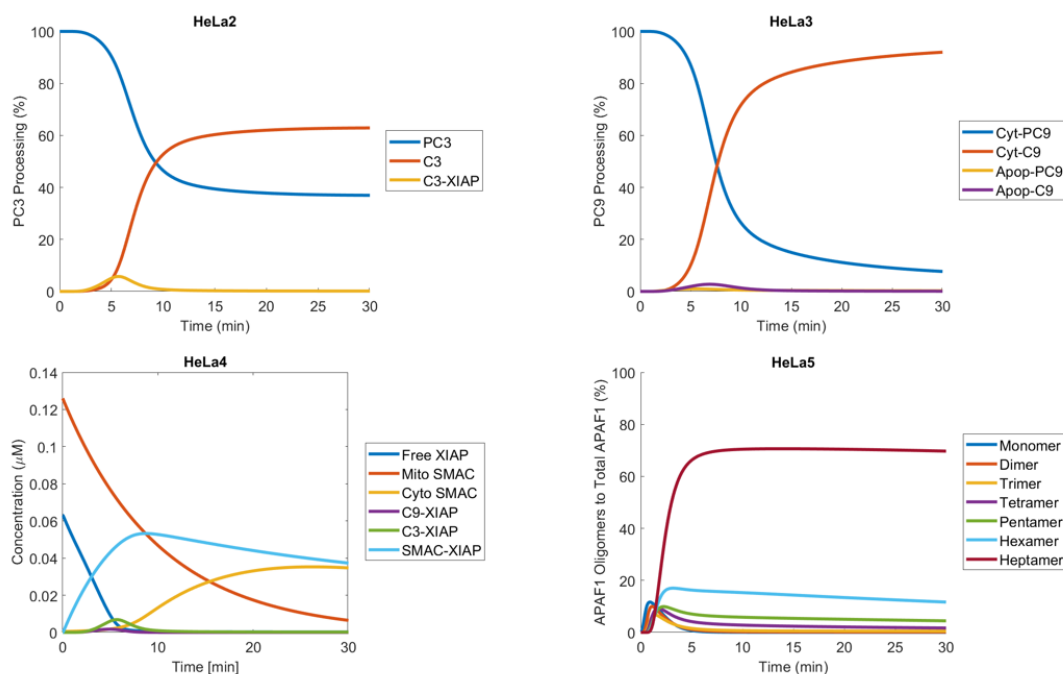


Figure 15: Temporal Development of Apoptotic Proteins in HeLa Cells Across Different Figure Modes.

9.4.3 Experiment 3

The simulation explores the effect of varying XIAP concentrations on the apoptosis pathway, ranging from 0 μM (knockout) to 0.40 μM (overexpression).

Key observations include:

- XIAP Loss:

When XIAP is completely absent (0 μM), there is a significant increase in caspase-3 activity. This leads to a higher rate of apoptosis, as XIAP normally acts as an inhibitor of caspase activity.

- XIAP Overexpression:

With increasing XIAP concentration, caspase-3 activity decreases, indicating reduced apoptosis. This is because XIAP binds to and inhibits caspases, preventing the execution phase of apoptosis.

- Comparison to Classical Enzyme/Inhibitor Relationships:

The observed trends are consistent with classical enzyme/inhibitor dynamics, where the presence of an inhibitor (XIAP) reduces the activity of the target enzyme (caspase-3).

- Cleaved and Free Caspase-3 Levels:

The levels of cleaved caspase-3, an active form, decrease with higher XIAP concentrations. Free caspase-3, the inactive form, increases due to the inhibitory effect of XIAP.

Interpretation of Results Based on Apoptotic Proteins:

- XIAP:

XIAP plays a critical role in regulating apoptosis by inhibiting caspases. The concentration of XIAP directly affects the balance between cell survival and death.

Low XIAP: Leads to enhanced apoptosis due to unrestrained caspase activity.

High XIAP: Suppresses apoptosis, highlighting XIAP's role as a key inhibitor in the apoptotic pathway.

- Caspase-3:

Caspase-3 is an essential executor of apoptosis. Its activity is modulated by XIAP, demonstrating the importance of XIAP in controlling apoptosis.

High Caspase-3 Activity: Occurs when XIAP levels are low, leading to rapid apoptosis.

Low Caspase-3 Activity: Results from high XIAP levels, reducing apoptosis.

- System's Sensitivity to XIAP:

XIAP concentration acts as a biological switch, where low XIAP levels lead to high caspase-3 activity and apoptosis, while high XIAP levels suppress apoptosis. This demonstrates the critical role of XIAP in regulating apoptosis.

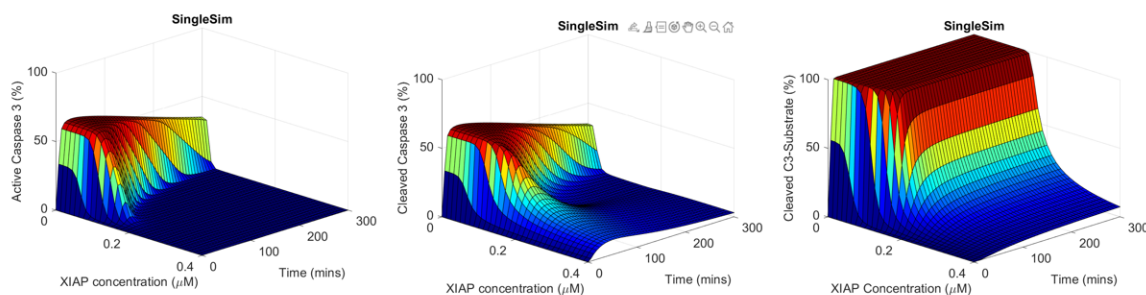


Figure 16: Effect of XIAP Concentration on Caspase-3 Activity in HeLa Cells

9.4.4 Experiment 4

The simulation investigates the effect of protein concentration variation on apoptosis in HeLa and HCT116 cells.

Observations:

- Mean Substrate Cleavage Kinetics:

For both HeLa and HCT116 cells, the mean substrate cleavage kinetics show a general trend, with individual traces deviating due to the imposed variation in protein concentrations.

- Increasing Standard Deviations:

As the SD increases from 10% to 30%, the spread of the individual traces around the mean becomes wider. This indicates greater variability in apoptosis outcomes due to increased noise in protein concentrations.

- Comparing HeLa and HCT116 Cells:

Differences between HeLa and HCT116 cells can be observed in their response to the variations. For instance, HeLa cells might show more pronounced fluctuations compared to HCT116 cells at higher SD levels, reflecting inherent differences in their apoptotic pathways. After 30 minutes, some traces in HCT116 cells do not reach 100% cleavage, suggesting that heterogeneity affects apoptosis sensitivity more significantly in HCT116 cells compared to HeLa cells. This indicates a higher impact of protein concentration variations on HCT116 cells.

Interpretation of Results Based on Apoptotic Proteins:

- Effect of Protein Concentration Variation:

Low Variation (10% SD):

Results are relatively consistent with the mean trace showing minimal deviations.

Moderate Variation (20% SD):

Greater deviations from the mean are observed, indicating that protein concentration noise begins to significantly impact the apoptotic response.

High Variation (30% SD):

The widest spread of individual traces is seen, highlighting the critical role of precise protein concentrations in determining apoptosis outcomes.

Transient vs. Stable Complexes:

- Stable Complexes:

These show minimal fluctuations across simulations, indicating robustness to protein concentration variations.

- Transient Complexes:

Significant fluctuations are observed, reflecting their sensitivity to changes in protein levels.

Differences Between Cell Types:

- HeLa Cells:

Display more pronounced responses to protein concentration variations, suggesting a more sensitive apoptotic network.

- HCT116 Cells:

Show relatively smoother responses, indicating a more stable apoptotic network under similar conditions.

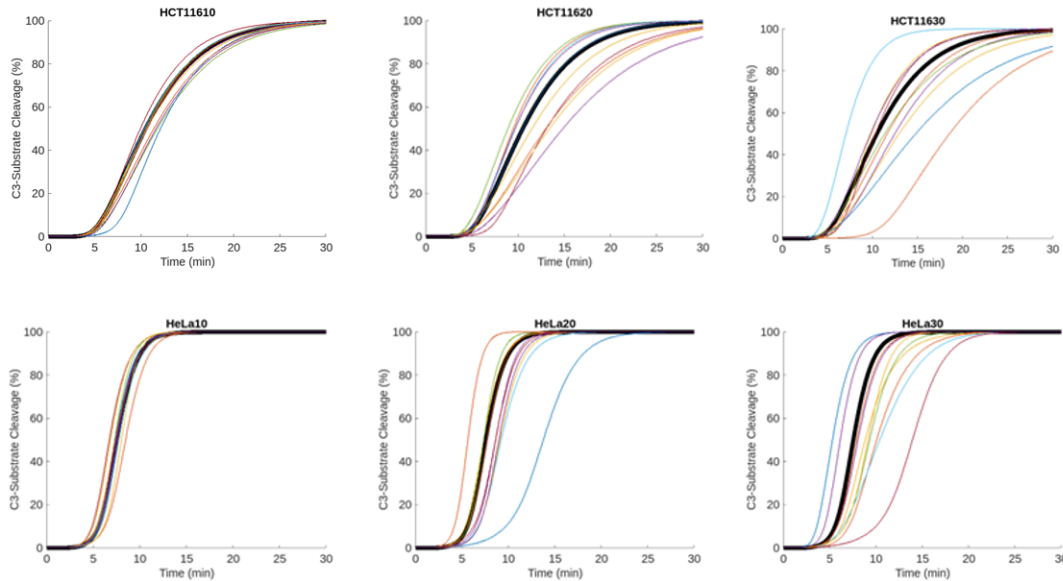


Figure 17: Time-Lapse Imaging

9.4.5 Experiment 5

The simulation separates patients into two groups based on their ability to cleave $> 25\%$ of caspase 3 substrate in 5 hours:

- Apoptosis Competent: Cleavage $> 25\%$
- Apoptosis Resistant: Cleavage $< 25\%$

Observations:

- Kaplan-Meier Survival Curves:

Patients in the apoptosis competent group show higher survival rates over time compared to those in the apoptosis resistant group.

- Logrank Test:

The p-value for the logrank test indicates a statistically significant difference in survival between the two groups.

- Hazard Ratio (HR):

The HR quantifies the risk of being apoptosis resistant vs. capable, with a higher HR indicating a greater risk of poor prognosis for apoptosis-resistant patients.

Interpretation of Results Based on Apoptotic Proteins:

- Impact of Caspase 3 Substrate Cleavage on Prognosis:

- Apoptosis Competent Group:

Higher cleavage rates of caspase 3 substrate suggest a more effective apoptotic response, leading to better patient outcomes.

- Apoptosis Resistant Group:

Lower cleavage rates indicate a deficient apoptotic response, correlating with poorer patient outcomes and higher mortality.

- Comparison with Classical Scenarios:

- Enzyme/Inhibitor Relationships:

The results align with classical scenarios where efficient substrate cleavage by enzymes (caspases) promotes apoptosis, while inhibition (e.g., by XIAP) reduces apoptosis.

- System-Level Behavior:

The differential survival outcomes underscore the critical role of apoptosis execution in patient prognosis, highlighting the importance of precise regulatory mechanisms in the apoptosis network.

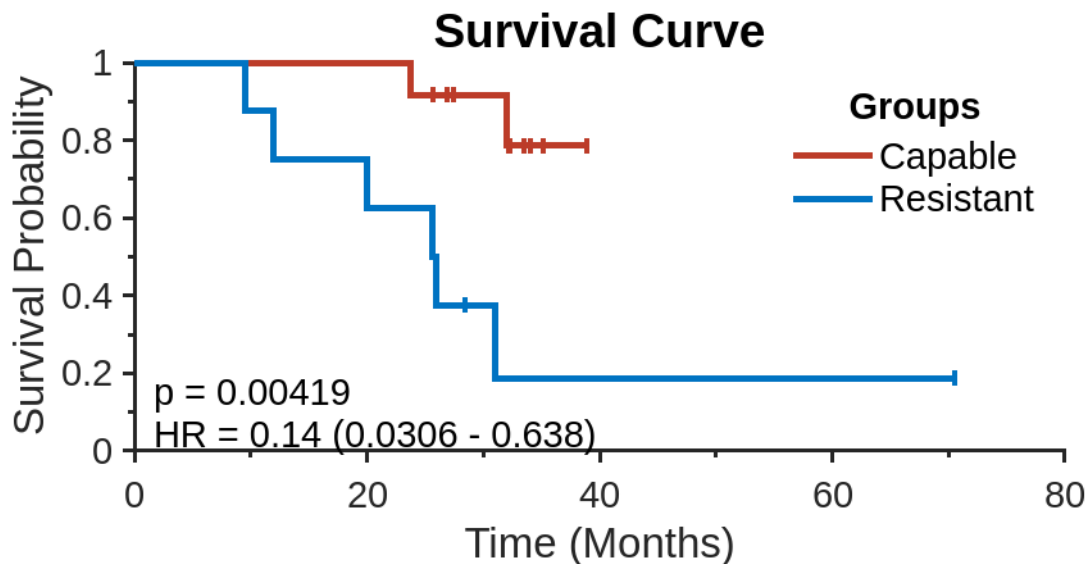


Figure 18: Kaplan-Meier Survival Curves for Colorectal Cancer Patients Based on Apoptosis Competency

9.5 Discussion

The simulations conducted in this experiment provide valuable insights into the kinetics and regulation of the apoptosis execution phase across different cell lines. Key observations include:

1. Differences in Apoptosis Kinetics: HeLa cells exhibited a more rapid and efficient apoptotic response compared to HCT116 cells, primarily due to higher initial concentrations of key apoptotic proteins and lower levels of inhibitors like XIAP. This highlights the variability in apoptotic signaling between different cell types.

2. Protein Interaction Dynamics: The transient and stable complexes formed during apoptosis play crucial roles in determining the pathway's efficiency. For example, stable complexes such as XIAP-CASP9 maintain apoptosis inhibition, whereas transient complexes like APAF1-initiator caspase complexes drive the dynamic progression of apoptosis.

3. Effect of XIAP Concentration: Variations in XIAP levels significantly impact caspase-3 activity and apoptosis outcomes. Lower XIAP concentrations lead to enhanced apoptosis, while higher XIAP levels suppress it, reflecting classical enzyme/inhibitor dynamics.

4. Impact of Protein Concentration Variability: Increased variability in protein concentrations leads to greater deviations in apoptosis outcomes. This underscores the importance of precise regulation and consistency in protein levels for reliable apoptotic responses.

5. Prognostic Implications: The Kaplan-Meier survival analysis demonstrated that colorectal cancer patients with apoptosis-competent profiles (higher caspase-3 substrate cleavage) had better survival rates compared to apoptosis-resistant patients. This finding underscores the potential of apoptosis competency as a prognostic marker in cancer treatment.

9.6 Conclusion

This experiment successfully utilized mathematical modeling to simulate and understand the apoptosis execution phase in different cell lines. The ApoptoEverything software provided a detailed analysis of the kinetics and interactions of key apoptotic proteins. Key findings include the differential apoptotic responses between HeLa and HCT116 cells, the significant impact of XIAP concentration on apoptosis regulation, and the prognostic value of apoptosis competency in colorectal cancer patients. These insights can inform future research and therapeutic strategies targeting apoptosis in various diseases.

9.7 References

- J.C. Butcher, 2000, Numerical methods for ordinary differential equations in the 20th century, ScienceDirect
M.A. Savitskaya et al., 2015, Mechanisms of apoptosis, Biochemistry Moscow 80

J. Jack et al., 2007 , Simulating Apoptosis Using Discrete Methods: a Membrane System and a Stochastic Approach, En Unconventional Computation
K. Schleich, I.N. Lavrik, 2013, Mathematical modeling of apoptosis, Cell Commun Signal